

Leveraging an October Surprise to Estimate Coattail Effects

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Abstract

Does a presidential candidate's popularity impact her party's down-ballot success? So-called "coattail effects" can be challenging to identify: it is difficult to distinguish between voters' opinion of a prominent candidate and her political party. To overcome this challenge, I exploit a shock to candidate popularity created by late-election information in the 2016 presidential election. Differences in states' early voting laws created variation in voters' ability to respond to FBI Director James Comey's letter to Congress. Results indicate that knowledge of the letter significantly increased Republican presidential vote share. Using an instrumented difference-in-differences model, I find that a one percentage point increase in Republican presidential vote share increases down-ballot Republican vote share by 0.79 percentage points.

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1 Introduction

Political parties act as a coordination mechanism through which voters are mobilized, resources raised, and policies passed in the name of advancing common ideological goals. They also lower informational costs to voters: rather than taking the time to evaluate every candidate each election, a voter can interpret partisan affiliation as a signal of a candidate’s values and objectives. This generates the possibility for electoral spillovers between candidates, especially from ones with high visibility to those with less. These spillovers are known as “coattail effects”: a candidate has coattail influence if down-ballot candidates in her same party receive votes they otherwise would not have received, save for her popularity [Miller, 1955]. Coattail effects, though widely believed to exist, can be challenging to quantify: the popularity of a single candidate is difficult to parse from the public’s sentiment toward her political party. In this paper, I provide the first causally identified estimates of presidential coattail effects.

I estimate presidential coattail effects using a natural experiment created by late-election information in the 2016 U.S. presidential election. On October 28, 2016, then-FBI Director James Comey sent a letter to Congress regarding Democratic presidential candidate Hillary Clinton. Comey announced the discovery of additional emails which pertained to an investigation into Clinton’s use of a private email server while serving as Secretary of State – an investigation he previously testified was complete. The letter reflected negatively on Clinton, but variation in early voting laws across states meant voters had differing abilities to react to the information. Many people in states with early voting had already cast their ballots. In my central identification strategy, I compare election outcomes in counties with no-excuse early voting available before 10/28 to those in counties where no-excuse early voting was either unavailable or began after 10/28, relative to election outcomes in 2012.

First, I show that the Comey letter impacted the 2016 presidential election: exposure to the letter decreased the Democratic presidential vote share by 2 percentage points and increased the Republican presidential vote share by 3.2 percentage points. This increase is larger than the vote margins of eight states in 2016: of these eight, four did not have no-excuse early voting ahead of October 28th, and three of those four (Florida, Michigan, and Pennsylvania) broke in favor of Republican candidate Donald Trump.¹ Using this as a first stage, I use an instrumented difference-in-differences model to estimate coattail effects. I find that a one percentage point increase in the Republican presidential vote share led to

¹Of the four that did have no-excuse early voting ahead of the letter’s release, three (Minnesota, Nevada, and Maine) broke in favor of Clinton.

a 0.79 percentage point increase in the average down-ballot Republican vote share.

The magnitude of these results is politically significant. The 2017-2019 Senate was comprised of 51 Republicans and 49 Democrats, with Vice President Pence casting 13 tie-breaking votes, more than all tie-breaking votes cast in the previous 14 Senates combined [Schaeffer, 2024]. Pence cast tie-breaking votes to confirm the nominations of the secretary of education and the deputy director of the Office of Management and Budget, as well as on measures related to healthcare, consumer protections, and Title X funding for family planning and related services [Ballotpedia, e].² Using my estimated effect of a 3.2 p.p. decrease in Democratic down-ballot vote share, 2 Republican seats in states most able to respond to the Comey letter would have been Democratic in the absence of the letter, flipping control of the Senate. As such, the coattail effects identified in this paper not only had political effects, but direct policy effects as well.³

We might be concerned that these findings reflect differential political trends of early voting and non-early voting states rather than the impact of the Comey letter. To address these concerns, I show that other items on states' ballots are unrelated to the timing and availability of early voting with a balance test. To add further support for the parallel trends assumption, I leverage several high-frequency datasets to compare treatment and control states in the weeks leading up to Election Day in 2012 versus 2016 with triple-difference event studies. I find no significant pre-trends using numerous measures of campaign activity and voters' intentions.⁴

To provide additional support for my results, I limit my analysis to six early voting states which separately report votes cast on Election Day versus during the early voting period.⁵ This allows me to compare partisan vote shares of early voters and Election Day voters to assess the Comey effect. These states had varying amounts of early voting available ahead of the letter's release. I expect to observe larger effects in states that had more (vs. less) days of early voting available prior to 10/28, as those states presumably had a larger proportion of voters casting their ballot with no knowledge of the letter. I use a difference-in-differences design with continuous treatment where the number of days of early voting available before 10/28 is the treatment dose. I find that in a county with 10.17

²Pence's tie-breaking vote to confirm Secretary of Education DeVos was the first instance of a Vice President using this power in a Cabinet nomination vote in U.S. history.

³For example, the Tax Cuts and Jobs Act was passed using the ballot reconciliation process, with 51 Republicans voting in favor and 49 Democrats voting against it. See https://www.senate.gov/legislative/LIS/roll_call_votes/vote1151/vote_115_1_00303.htm.

⁴These exercises are all contained in the Online Appendix.

⁵The six states are Arkansas, Georgia, Iowa, Louisiana, Maryland, and North Carolina.

days of early voting ahead of the Comey letter (the average number of days of early voting prior to 10/28 across all states with no-excuse early voting in 2016), exposure to the letter decreased Democratic presidential and down-ballot vote shares by 2.6 and 2.8 percentage points, respectively, and increased Republican presidential and down-ballot vote shares by 1.7 and 1.9 percentage points, respectively. These findings stem from a different source of variation but are comparable to those from my main specification.

To better understand the mechanisms behind these results, I provide a stylized model of coattail effects with a negative shock to candidate valence. Formal theoretical treatments of coattail effects include Zudenkova [2011] and Halberstam and Montagnes [2015]. Zudenkova [2011] shows that such effects arise if we assume incumbents prefer their co-partisans to win races in different levels of government.⁶ Halberstam and Montagnes [2015] develop a model in which uninformed voters are assumed to be straight-ticket voters who align their down-ballot vote choice with their presidential one. These are low-propensity voters who vote in presidential elections but not midterm elections, generating ideological differences in senators elected in presidential and midterm years.⁷

I focus on the case of presidential elections in Halberstam and Montagnes’ model, adapting it to allow uninformed voters to abstain. Assuming that voters derive expressive utility from candidates’ ideology and valence, these voters may respond to negative information about their ideologically-preferred candidate by voting for her opponent or abstaining entirely. I demonstrate how the Comey letter could have generated coattail effects via two distinct mechanisms. First, the letter may have decreased the proportion of uninformed voters who preferred the Democratic presidential candidate (the “vote-switching” channel). Second, the letter may have increased the proportion of uninformed voters who preferred the Democratic presidential candidate but chose to abstain (the turnout channel).

I formally test the turnout mechanism using daily counts of early votes cast by registered party in North Carolina. This unique system of reporting early voting turnout provides for a natural comparison between pre-October 28th and post-October 28th early voters in 2012 versus 2016 with a two-way fixed effects event study. I find that the letter decreased

⁶Knowing this, retrospective voters (meaning they vote based on an incumbent’s performance) jointly evaluate co-partisans, providing for two-sided coattails (that is, spillovers both up and down the ballot).

⁷The authors document that senators first elected in presidential years are more ideologically extreme than those first elected in midterms and devise the model to explain this observation. It should be noted that this pattern of differing senator ideologies may have other explanations. For example, it could be that parties run more extreme Senate candidates to generate additional excitement for the party in presidential years. The documented correlation is between senator ideology and the *presence* of a presidential election, not co-partisan presidential success, and thus could be a result of strategic choices by both parties in presidential years, independent of their presidential candidates’ popularity.

registered Democrats’ share of early turnout by 2.2 percentage points in North Carolina. I find similar results using weekly early voting turnout data from Nevada and survey data on turnout intentions from the Cooperative Congressional Election Study. These findings provide evidence that the Comey letter generated coattail effects for down-ballot Democrats by decreasing turnout among those who otherwise would have voted for them.

This is the first paper to leverage an October surprise to estimate coattail effects. Although there is a large literature devoted to estimating coattail effects, proper identification of presidential coattails is difficult to achieve. The strategy used here offers a new method of isolating a shock to top-ballot candidate popularity and allows for formal testing of the mechanisms behind coattail effects. Additionally, I provide the first quasi-experimental analysis of the Comey letter’s impact on the 2016 presidential election and concurrent down-ballot races. These findings are politically significant and have direct policy implications given the narrow Republican control of the 2017-2019 Senate. My findings imply that coattail effects can lead to more instances of unified government, which may affect both legislative efficacy and the types of policies passed.

The presence and magnitude of coattail effects also have significant strategic implications for politicians. While coattail effects are likely not a determinant of down-ballot candidates’ party affiliation,⁸ they may influence how closely such candidates choose to align themselves with up-ballot ones. Presidential coattail effects may even influence *when* down-ballot candidates run: they may wish to obtain an electoral boost from a popular co-partisan or avoid being dragged down by an unpopular one. Finally, this paper’s finding that late-election information can impact presidential and down-ballot races suggests that the incentives for the strategic release of such information may be even larger than previously thought [Gratton et al., 2018].

2 Background

2.1 Coattail Effects

Though there is a large literature on coattail effects, credible identification is difficult: there are inherent complications in separating a particular candidate’s popularity from that of

⁸Though there is a recent uptick in party-switching among state lawmakers in the U.S. [Crampton, 2023].

her political party.⁹ How might we distinguish coattail effects from party politics? Coattail effects are defined as the impact of a candidate’s personal popularity on her co-partisans’ success elsewhere on the ballot. Thus, proper identification of such an effect requires some exogenous shock to a candidate’s popularity – importantly, separate from the favorability of her party at large – and variation in voters’ experience of that shock.

Meredith [2013] exploits the excess support granted by (geographically) nearby voters to estimate gubernatorial coattail effects, finding that a one percentage point increase in gubernatorial vote share leads to 0.1-0.2 percentage point increases in vote shares for partisan allies running for secretary of state and attorney general. Many recent papers use close-election regression discontinuity designs to estimate the effect of a co-partisan winning a prior election in a different level of government. These have yielded mixed results on the existence and direction of such “incumbency spillovers,” depending on setting, the levels of government, and the direction [Hainmueller and Kern, 2008, Broockman, 2009, Folke and Snyder, 2012, Ade and Freier, 2013, Feierherd, 2020, Ventura, 2021, Savu, 2024].¹⁰

It is possible that each of these designs could be capturing not just an increase in the personal popularity of a candidate, but also an amplified opportunity for party rhetoric. Geographically proximate candidates and incumbents alike have increased opportunities to positively advertise their own political party, complicating the distinction between coattail effects and party politics. Instead, the Comey letter (or more broadly, late-election information) provides a plausibly exogenous shock to candidate popularity alone.

Late-election information offers the popularity shock needed to estimate presidential coattails, and disparate state-level early voting laws supply the necessary variation. Key to my empirical design is the notion that convenience voters and Election Day voters face fundamentally different information environments [Meredith and Malhotra, 2011]. My identification strategy is similar to those used in Montalvo [2011], which shows that a terrorist bombing in Spain negatively impacted the performance of the incumbent party in the 2004 general election which took place three days later, and Graham and Svolik [2020], which demonstrates that a Montana U.S. House candidate’s assault of a journalist the night before

⁹Most prior work consists of observational studies which attempt to control for determinants of down-ballot vote shares [Campbell and Sumners, 1990, Mondak, 1993, Flemming, 1995, Mattei and Glasgow, 2005] or structural models [Kramer, 1971, Ferejohn and Calvert, 1984] which have proven to be sensitive to model specification [Fair, 2009]. See Meredith [2013] for a detailed discussion of the difficulties of identifying coattail effects.

¹⁰Several of these papers estimate “reverse coattails,” which refer to spillovers from down-ballot candidates to their fellow party members higher up on the ticket; for example, from a Democratic mayoral candidate to a Democratic U.S. House candidate.

an election hurt his performance with moderate voters on Election Day. This paper is the first to use such a design to 1) analyze the impact of late-election information in a U.S. presidential race and 2) identify coattail effects.

The paper is also related to work on the impact of information and media in elections [Couttenier et al., 2024, Ash and Galletta, 2023, Wang, 2021, Morton et al., 2015, Chiang and Knight, 2011, DellaVigna and Kaplan, 2007]. Some work has been done on the impact of the Comey letter, specifically: there is some evidence that the letter had impacts in electoral prediction markets [Halcoussis et al., 2020]. McKee et al. [2019] points out that in each of the seven swing states that Trump won, he lost the sum of the votes cast early in-person or by mail; this paper builds on this by using variation in early voting to provide the first quasi-experimental analysis of the Comey letter.

Coattail effects can have important effects on voter welfare through increasing the likelihood of unified government: a popular president-elect is more likely to take office with partisan control of both congressional chambers. This may make voters worse off if they prefer that executive and legislative branches are controlled by opposing parties to moderate policy outcomes [Alesina and Rosenthal, 1996] or provide more effective oversight of the executive [Fox and Van Weelden, 2010]. At the same time, unified government may decrease legislative gridlock: Tsebelis [1995] shows that the likelihood of policy reform is decreasing in the number of political veto players. Indeed, studies find that under divided governments, budget approvals are delayed [Kirkland and Phillips, 2018] and legislators’ ability to respond to revenue shocks is reduced [Alt and Lowry, 1994]. Empirical evidence on policy outcomes under divided governments is mixed: some studies have found that state legislators under divided governments implement more moderate policies [Repetto and Sosa Andrés, 2023] or spend more on bipartisan priorities such as infrastructure [Buchheim and Fretz, 2020]. However, Bernecker [2016] finds that in the US, state legislatures under divided governments are 25% *more* likely to adopt a welfare reform, theorizing that when state legislatures are controlled by the governor’s opposition party, they can confront the governor with policy issues through passing bills. The large documented effects of unified versus divided governments, in combination with the magnitude of estimates found in this paper, highlight the importance of coattail effects in determining policy.

2.2 Convenience Voting in the United States

Figure 1 displays statewide variation in no-excuse convenience voting in 2016. Thirty-five states began convenience voting before October 28th, 2016, the day of the Comey letter’s

publication. A total of thirty-seven states practiced no-excuse convenience voting at the time: note that Florida and Oklahoma had no-excuse convenience voting, but their early voting periods did not open until *after* October 28th. Of those thirty-seven, three states used universal vote-by-mail, meaning ballots were mailed to all registered voters.¹¹ The rest of the thirty-seven had early in-person voting options, where citizens could cast their ballot in-person up to fifty days ahead of Election Day.¹²

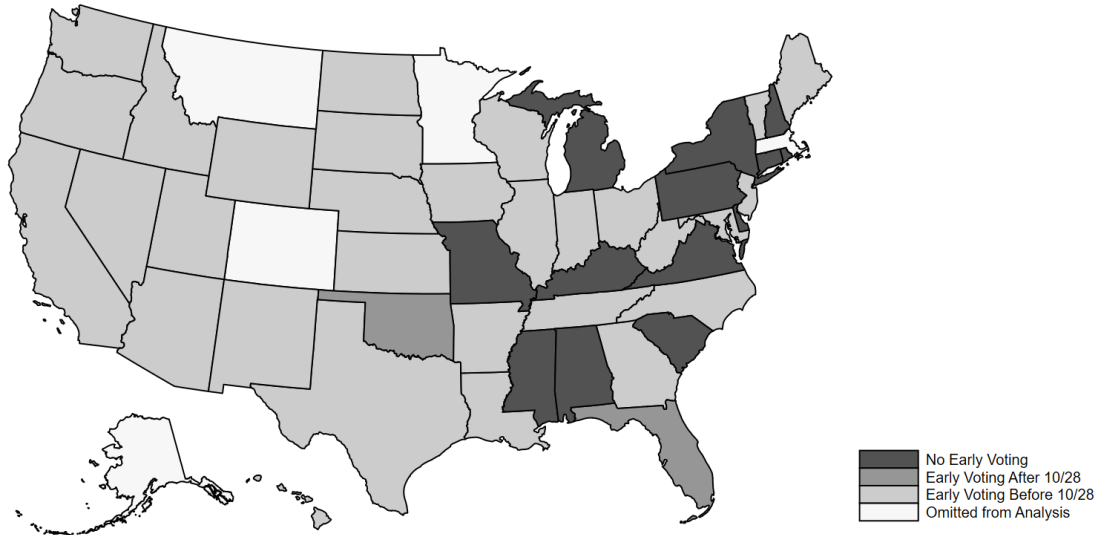


Figure 1: This map displays variation in convenience voting laws for U.S. states in 2016, using categorizations from [Ballotpedia](#) [d].

Notes: Four states (Colorado, Massachusetts, Minnesota, and Montana) made changes to their convenience voting laws between 2012 and 2016; I omit them from my analysis. I also omit Alaska, as it does not report election results by county. The map was created using “spmap” and “geo2xy” in Stata [Pisati, 2007, Picard and Stepner, 2017].

See Table 1 for summary statistics on counties in states with convenience voting and states without convenience voting.¹³ A potential concern with comparing the two groups is that there may be underlying qualities which differ between the two and are somehow correlated

¹¹These three states were Colorado, Oregon, and Washington; nearly 80% of counties in Utah were also using universal vote-by-mail in 2016 [Thompson et al., 2020].

¹²The earliest early voting start date in 2016 was September 19th in Wisconsin Rapids, Wisconsin.

¹³Counts of 18+ population are from the Surveillance, Epidemiology, and End Results Program (SEER) [National Bureau of Economic Research (NBER), Surveillance, Epidemiology, and End Results Program (SEER)]. Political variables (vote margin, vote share, and turnout) are from Dave Leip’s Atlas of U.S. Presidential Elections [Leip]. I obtain county-level control variables from the U.S. Census Bureau’s American Community Survey 5-year estimates, the U.S. Bureau of Labor Statistics, and the U.S. Census Bureau’s Census of Population and Housing.

with treatment. That is, if one group was more or less responsive to the Comey letter due to differential baseline characteristics. However, as Table 1 shows, the two groups possess relatively comparable baseline demographic, economic, and political attributes.

Table 1: This table displays summary statistics in 2012 for counties in the United States, broken into states with convenience voting and counties without convenience voting.

	Convenience voting	No convenience voting	Difference (std. devs.)
18+ population	78,171.433	77,744.844	0.002
Unemployment rate	7.554	8.740	0.424
Population density	176.719	519.702	0.188
Median age	40.258	40.279	0.004
Number of men per 100 women	100.827	98.334	0.221
Share non-white	15.434	18.461	0.183
Median income (in \$10,000s)	4.554	4.411	0.122
Share with Bachelor’s or higher	19.024	19.148	0.015
Share white without college degree	79.907	79.267	0.070
Dem./Rep. House margin*	-0.233	-0.168	0.187
Dem. vote share in pres. race	0.368	0.417	0.336
Turnout	0.549	0.568	0.193
Count	2090	801	

Notes: The means of the convenience voting and no convenience voting groups are reported, along with the difference in terms of the standard deviation of the entire sample, for reference. For example, the difference in 18+ population between the groups is 0.002 of one standard deviation (measured from the entire sample). The variable denoted with an asterisk is averaged over three U.S. election cycles: 2008, 2010, and 2014. All other variables are taken from 2012. Colorado, Massachusetts, Minnesota, and Montana are not included, as they changed convenience voting laws between 2012 and 2016; Alaska is not included, as it does not report election results by county. Population density is calculated by dividing a county’s 18+ population by its land area in square miles (as reported by the 2010 Census). The share with Bachelor’s or higher variable refers to the percent of those aged 25+ with a bachelor’s or more advanced degree.

2.3 The Comey Letter as Late-Election Information

Late-election information, broadly speaking, is revealed to voters in the weeks or days preceding Election Day and has the potential to change voters’ minds. This information can take the form of something already structured within the political system, such as presidential debates or campaign stops, or a spontaneous revelation, known as the “October surprise” in American politics. Interested parties certainly have incentives to release information with the intent to impact election outcomes as Election Day looms.¹⁴ However, this late-election

¹⁴Gratton et al. [2018] present this as a tradeoff between credibility and scrutiny: good signals of candidate quality are released earlier, as this conveys credibility of the information, but poor signals will be released later, since they would not hold up to the public’s scrutiny if examined for too long.

information, whether spontaneous or strategically released, can nevertheless “surprise” the public and might induce changes in voter behavior.

There are many recent examples of late-election information in the United States: in 2020, news of an extramarital affair and a subsequent investigation by the U.S. Army Reserve of Cal Cunningham, a Democrat running for a U.S. Senate seat in North Carolina, occurred less than one month away from Election Day [Robertson, 2020]. In the 2003 California gubernatorial recall election, Republican Arnold Schwarzenegger replaced incumbent Democratic Governor Gray Davis, winning by a margin of 968,491 votes [Ballotpedia, f]. The *Los Angeles Times* reported on the Friday before the election that six women had accused Schwarzenegger of sexual misconduct [Cohn et al., 2003] after “more than three million of ten million voters had already cast their ballots” [Gronke et al., 2008]. In 2000, news of George W. Bush’s 1976 DUI arrest was reported on by media outlets four days before Election Day [Balz, 2000].

In this paper, I focus on a particular example of late-election information: then-FBI Director James Comey’s October 28, 2016 letter to Congress. Comey announced in the letter that the FBI had discovered more emails which seemed pertinent to the investigation of Hillary Clinton’s use of a private email server during her time as Secretary of State (an investigation Comey had previously testified was complete). The letter was sent to Congress just eleven days before the presidential election between Republican candidate Donald Trump and Democratic candidate Hillary Clinton [Perez and Brown, 2016].

The last month or so of the campaign leading up to Election Day on November 8th was quite dynamic in terms of late-election information. On October 7th, *The Washington Post* obtained a video in which “Donald Trump bragged in vulgar terms about kissing, groping and trying to have sex with women during a 2005 conversation caught on a hot microphone” [Fahrenthold, 2016]. This prompted several prominent Republicans to condemn Trump; some even suggested he withdraw his candidacy [Wellford, 2016]. On that same day, the Office of the Director of National Intelligence and the Department of Homeland Security released a statement accusing Russia of interfering with the election, and media organization WikiLeaks began releasing emails belonging to Clinton campaign chairman John Podesta, many of them “embarrassing for Clinton” [Cohen, 2017]. These releases continued on a near-daily basis throughout the rest of the campaign, but none seemed to generate a nation-wide shock in the way that 1) the aforementioned tape, 2) the beginning of the WikiLeaks releases, or 3) the Comey letter did.

I choose to focus on the Comey letter as the late-election information of interest in the election. The events of October 7th, 2016, negatively impacted both candidates, prompting muddled behavioral implications for voters. In comparison, the Comey letter (and thus October 28th) clearly represented negative information for Clinton alone. Additionally, not as many people had yet had the opportunity to vote by October 7th, making for a smaller control group: just ten states began convenience voting before October 7th.¹⁵

As mentioned above, thirty-five states began convenience voting before October 28th. Although concerns over Clinton’s use of the email server had already been raised and an investigation completed, this letter and its implication of a re-opened investigation may have legitimized some voter’s concerns about the candidate. It certainly represented an information shock of interest to many voters: Figure 2 displays Google Trends data, which shows a large spike in searches for “Comey” on 10/28/16 [Google Trends]. Election Day was still eleven days away, yet many had already cast their ballot before they learned of this investigation. I leverage this variation in voters’ knowledge of the letter at the time of casting their ballot, plausibly exogenous due to differences in states’ early voting policies, to provide causally identified estimates of the letter’s effect on election outcomes in 2016.

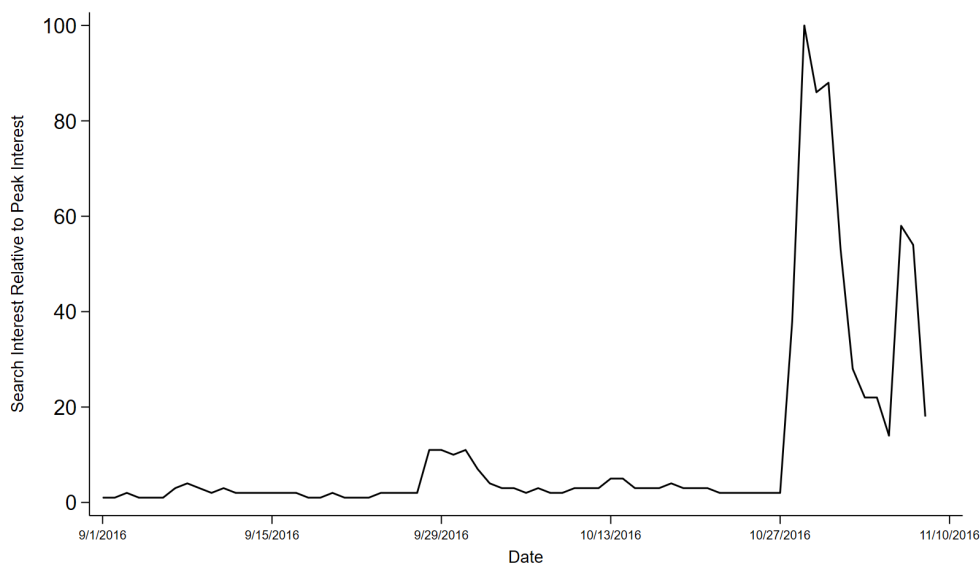


Figure 2: This figure shows Google Trends data for searches for “Comey” in the United States between 9/1/2016-11/8/2016. A value of 100 is peak popularity for the term.

¹⁵In a robustness check, I show that results in this paper do not hold if treatment is redefined around early voting availability ahead of October 7th, emphasizing the conflicting effects the numerous events of the day may have had on voters: see Online Appendix Table 5.

3 Theoretical Framework

In the following section, I adapt the model of coattails developed by [Halberstam and Montagnes \[2015\]](#) to highlight two mechanisms through which the Comey letter may have affected down-ballot candidate performance: vote-switching and changes in turnout. First, I summarize the setup of their model.¹⁶ Assume there are two contemporaneous races (here, presidential and down-ballot – this could be a Senate race, a U.S. House race, gubernatorial election, etc.). Two candidates (a Democrat and a Republican) compete in each race. Suppose that candidates’ ideologies are independent of one another: in particular, Democratic presidential and down-ballot candidates’ ideologies are independent draws from a uniform distribution from -1 to 0 , and Republican presidential and down-ballot candidates’ ideologies are independent draws from a uniform distribution from 0 to 1 . There are many citizens, each of whom have some ideological bliss point, distributed uniformly between -1 and 1 .

There are two types of citizens: informed voters, who observe the ideological positions of all four candidates, and uninformed voters, who observe only the ideological positions of the presidential candidates. Let uninformed voters constitute a proportion $\delta > 0$ of the electorate. Suppose for now that if voters know the ideological positions of the candidates, they vote for the candidate they are ideologically closer to. Uninformed voters are assumed to vote for the down-ballot co-partisan of their preferred presidential candidate. That is, if an uninformed voter votes for the Democratic presidential candidate, she votes for the Democratic down-ballot candidate, too. This assumption is grounded in the idea that uninformed voters are more likely to be low-propensity voters [[Halberstam and Montagnes, 2015](#)]; one explanation for this is that they may have higher informational costs associated with voting. Therefore, they are more likely to rely on partisan affiliation and (more easily accessible) information about the top-ballot candidate to determine their down-ballot vote choice.

Suppose for now that all citizens vote. Let π be the proportion of uninformed citizens that prefer the Democratic presidential candidate (and therefore will vote for the Democratic down-ballot candidate). Then, the Democratic down-ballot candidate wins if and only if

$$\frac{y_D + y_R}{2} > (1 - 2\pi) \frac{\delta}{1 - \delta}, \quad (1)$$

where y_D and y_R are the ideological positions of the Democratic and Republican down-ballot

¹⁶Microfoundations of Halberstam and Montagnes’ model can be found in the Online Appendix of their paper.

candidates.¹⁷

To understand how the Comey letter might have led to negative electoral spillovers for down-ballot Democrats through vote-switching or changes in turnout, we must introduce both candidate valence and the issue of political participation. Assume that voters' preferences over presidential candidates depend on their ideological proximity to the candidates as well as candidate valence, as in the model of expressive voting in bundled elections developed by Knight [2017].¹⁸ Voters turn out if and only if the total expressive benefits from voting in each race (presidential and down-ballot) exceed the cost of doing so.¹⁹ Conditional on turning out, voters select presidential candidates based on both ideology and candidates' relative valence. Continue to assume that informed voters base their down-ballot vote on their ideological proximity to the candidates and that uninformed voters align their down-ballot vote with their presidential one.

While all voters value candidate valence, relatively moderate voters are the most likely to respond to a valence shock, as was brought on by the Comey letter.²⁰ Again, voters turn out so long as the expressive benefits of voting for each candidate exceed the cost of voting. For voters with strong partisan ties, i.e. ones whose ideological bliss points lie definitively on the left or on the right, ideological expressive utility alone is likely sufficient to lead them to turn out. It would take quite a large difference in candidate valence to cause them to abstain, and an even more extreme one to lead them to vote for the candidate whom they are ideologically further from.

Voters with more moderate ideological bliss points, on the other hand, are much more likely to be impacted by a valence shock. First, they are less likely to vote in the first place. Since their utility is less impacted by the ideological difference between candidates, they require a sufficiently large difference in candidate valence to turn out. Second, for that same reason, their vote choice is much more easily swayed by valence shocks. That is, a moderate left-leaning voter, upon learning of the Comey letter, might have responded by either 1) deciding to abstain, if the decline in relative valence for Clinton leaves her expressive utility lower than the cost of voting, or 2) deciding to vote for Trump, if the difference in candidate

¹⁷Note: this is equation (1) in Halberstam and Montagnes [2015].

¹⁸To focus on the impact of presidential candidates' valence, I choose not to assign valence characteristics to down-ballot candidates, similar to the assumption that House candidates are of equal quality in Knight [2017]. Given that uninformed voters do not observe down-ballot candidates' policy positions, it is plausible they would not be aware of down-ballot candidates' quality anyway.

¹⁹In particular, they derive utility from expressing their preferences over each set of candidates.

²⁰Throughout this section, consider a "valence shock" to be a change in candidates' relative valence, i.e. the difference in candidate quality.

valence is large enough such that her expressive utility of voting exceeds the cost.²¹

Halberstam and Montagnes posit that informed voters likely have more extreme ideological positions than uninformed voters (and document this with survey data). As such, informed voters are likely unaffected by valence realizations. Here, I will continue to make the simplifying assumption that all informed voters turn out, and I will further assume that informed voters have sufficiently extreme ideological preferences such that their presidential vote choice is unaffected by valence.²² However, uninformed voters, who likely have more moderate policy preferences, may choose to switch their vote or abstain entirely in the presence of a negative valence shock for their ideologically-preferred presidential candidate as described above. Given that uninformed voters are straight-ticket voters, a Democratic presidential candidate's valence shock can affect the down-ballot Democrat, who loses voters through decreased turnout and/or vote-switching.

First, consider the vote-switching mechanism. To isolate this mechanism, assume that the cost of voting is zero, so that all voters turn out.²³ Note that the right-hand side of condition (1) is decreasing in π , the share of uninformed voters that prefer the Democratic presidential candidate.²⁴ If the Democratic presidential candidate experiences a negative valence shock, then some uninformed voters who are ideologically closer to the Democratic presidential candidate will now prefer her Republican opponent (vote-switching), decreasing π . As shown by Halberstam and Montagnes, for a given y_R , the probability the Democratic down-ballot candidate wins is $\min\{y_R - \frac{2\delta}{1-\delta}(1 - 2\pi), 1\}$, which is increasing in π . Therefore, as π decreases from the negative valence shock to the Democratic presidential candidate, the likelihood that the Democratic down-ballot candidate wins also decreases.

Next, let voting costs be non-zero and consider the turnout mechanism. Now, suppose that a proportion ρ_D of uninformed voters who prefer the Democratic presidential candidate turn out (define ρ_R similarly for those who prefer the Republican candidate).²⁵ Then, the

²¹Additionally, the Comey letter might have led some moderate right-leaning voters who were previously planning to abstain to decide instead to turn out and vote for Trump, as it would have increased their expressive utility stemming from the presidential candidates' relative valence.

²²That is, I assume the cost of voting is sufficiently low and consider only modest shocks to candidate valence, which are not large enough to affect the behavior of those with strong partisan ties. These assumptions could be relaxed, but I am interested in the dynamics generated by changes in behavior among uninformed, low-propensity voters.

²³If a voter is indifferent between candidates, assume they decide with a coin flip.

²⁴Now, this share reflects preferences based on ideology and valence, rather than ideological proximity to the presidential candidate alone.

²⁵Those whose expressive utility of voting outweighs the cost will choose to turn out.

Democratic down-ballot candidate wins if and only if

$$\frac{y_D + y_R}{2} > \frac{\delta}{1 - \delta}(\rho_R - \pi(\rho_R + \rho_D)), \quad (2)$$

where the right-hand side is decreasing in ρ_D . For a given y_R , the probability the Democratic down-ballot candidate wins is thus $\min\{y_R - \frac{2\delta}{1-\delta}(\rho_R - \pi(\rho_R + \rho_D)), 1\}$, which is increasing in ρ_D . Therefore, as ρ_D decreases in response to a negative valence shock for the Democratic presidential candidate (as it would through the turnout mechanism: $\rho_D = \rho_R = 1$ without abstentions, yielding condition (1)), the likelihood that the Democratic down-ballot candidate wins decreases.²⁶

From conditions (1) and (2), we can see how the Comey letter may have generated coattail effects for down-ballot Democrats in 2016. In particular, uninformed voters, who are more likely to be ideologically moderate [Halberstam and Montagnes, 2015], are assumed to be straight-ticket voters who align their down-ballot vote with their presidential one. Moderate voters respond to negative valence shocks to their ideologically-preferred presidential candidate by either voting for her opponent or abstaining entirely, creating spillover effects for down-ballot Democrats through lost votes due to vote-switching and/or decreases in turnout.²⁷

It should be noted that since I theorize that the letter's effects were driven by changes in turnout or vote-switching by uninformed, low-propensity voters, the findings in this paper may not be generalizable to coattails generated by top-ballot candidates in midterm elections, where low-propensity voters are much less likely to turn out. At the same time, if a top-ballot candidate in a midterm year is popular enough to mobilize low-propensity voters, one could imagine positive spillovers for her down-ballot co-partisans. Further research is needed to understand the mechanisms which generate coattail effects in midterm elections.

²⁶The share of voters who prefer the Democratic presidential candidate, π , would still decrease in this case. Similarly, the proportion of uninformed voters who prefer the Republican candidate who choose to turn out, ρ_R , would likely increase. Both of these would serve to further decrease the likelihood the Democratic down-ballot candidate wins.

²⁷If we considered a valence shock extreme enough to affect the behavior of informed voters, informed voters could generate coattails through the turnout mechanism, but not the vote-switching one. This is because they do not align their down-ballot vote with their presidential one, and thus, conditional on an informed voter continuing to turn out, her down-ballot vote would be unaffected by changes in presidential candidates' relative valence.

4 Main Analysis

4.1 State-Level Variation in Convenience Voting

4.1.1 Empirical Approach

In my main empirical strategy, I leverage cross-state variation in the availability of early voting. I obtain data on convenience voting laws and timing from the Elections & Voting Information Center for the 2012 election [[Elections & Voting Information Center, 2012](#)] and Ballotpedia for the 2016 election [[Ballotpedia, d](#)] and use it to define treatment in a difference-in-differences design. I categorize two types of counties as treated: counties in states without no-excuse early voting and counties in states where early voting did not open until after October 28th, 2016. The treatment turns on with the Comey letter (only in 2016 and not in 2012). I use counties in states with early voting which opened before October 28th, 2016 as control units, as voters in these states would have been least able to respond to the information. Here, the variation of interest comes from the availability of no-excuse early voting, something set independently by each state.²⁸

I use county-level election returns to create the outcomes of interest: turnout rate (total ballots cast / voting-age population), vote shares for Democratic and Republican presidential candidates, and average vote shares for Democratic and Republican down-ballot candidates [[Leip](#)].²⁹ To determine partisan vote shares for down-ballot candidates, I average together the Democratic (or Republican) vote shares in U.S. House, Senate, and gubernatorial elections.³⁰ The identifying assumption here is that, absent the Comey letter, treatment and control counties would have trended similarly in the outcomes of interest. This design also requires that any late-election information in 2012 did not affect voting behavior.³¹ I estimate

$$y_{ct} = \alpha + \delta T_c + \gamma P_t + \beta(T_c \times P_t) + \epsilon_{ct}, \quad (3)$$

²⁸I exclude from my analysis any state which changed its convenience voting laws between 2012 and 2016 (Colorado, Massachusetts, Minnesota, and Michigan).

²⁹Counts of voting-age population (i.e., 18+) come from [[National Bureau of Economic Research \(NBER\), Surveillance, Epidemiology, and End Results Program \(SEER\)](#)].

³⁰I exclude races without both a Democratic and Republican candidate running, as vote shares would not necessarily reflect any spillover effects in these cases.

³¹Identification requires that states' early voting laws do not change between reference elections and 2016, so that any results are attributable to the Comey letter and not changes in convenience voting opportunities. I opt to use only 2012 as a comparison year to keep the sample of states as large as possible, as well as to satisfy the requirement that reference elections not feature October surprise(s) near October 28th.

where T_c takes a value of 1 for counties in states without early voting and states where early voting did not open until after October 28th, 2016 and 0 otherwise, and P_t takes a value of 1 in 2016 and 0 otherwise. I weight regressions by county-year voting-age population, as is common in the literature [Charles and Stephens Jr., 2013, Brunner et al., 2011]. I cluster at the state level, as that is the level at which treatment (early voting availability) is assigned.

4.1.2 Support for Identification

One potential threat to identification with the comparison of early voting to non-early voting states is the possibility that these two sets of states may have been experiencing different political trends. If this were true, we might expect to see it directly reflected in those states' ballots. Perhaps one set of states had a separate reason to swing left or right in 2016: toss-up down-ballot races, ballot initiatives, etc. I collect data on states' 2012 and 2016 ballot characteristics in four broad categories: toss-up Senate or gubernatorial races [Cook Political Report, 2016c,b, 2012c,b], the number and type of ballot initiatives [Ballotpedia, c,b], third-party presidential candidates on the ballot [Leip], and straight-ticket voting [The National Conference of State Legislatures].^{32,33} I also collect data on states' "battleground status," as states expected to have especially competitive presidential races may have drawn outsized campaign activity [Cook Political Report, 2016a, 2012a].³⁴ Results from a balance test using these variables are shown in Online Appendix Figure 1; I find that there is no relationship between treatment and any of these items on the ballot or battleground status.

Another way to test for differing political trends is to examine if campaign activity differed across treatment and control states. For example, if campaign spending were differentially high in treated states in 2012 or 2016, this would pose an issue for identification. I collect data on total campaign spending and campaign donations to candidates in 2012 and 2016 [Federal Election Commission]. As an additional measure of campaign activity, I use data from the Wesleyan Media Project on total campaign advertisements favoring down-ballot Democratic or Republican candidates in 2012 and 2016 [Franklin Fowler et al., 2023, Fowler

³²Toss-up races are the most competitive races of the election cycle, where neither party is predicted to have a slight edge.

³³Straight-ticket voting allows a voter to vote for all of their chosen party's candidates for each partisan office on the ballot with a single mark of their ballot.

³⁴I categorize all states deemed to have competitive presidential races by Cook Political Report as battleground states.

et al., 2023].³⁵ I aggregate these data to the state-week level, allowing me to look at trends throughout the election cycles in 2012 and 2016. I estimate separate two-way fixed effects models for 2012 and 2016 and implement a triple-difference strategy to compare the two years directly. Results are shown in Online Appendix Figures 2, 3, and 4. There are no significant pre-trends in (logged) campaign expenditures or donations, and differences in these measures cannot explain my results. In 2016, there were increased campaign ads in non-early voting states in the very early weeks of the election, but this did not differ significantly by party (see Online Appendix Figure 5).

To add further support for the assumption of parallel trends, I turn to prediction market prices to capture estimates of election outcomes throughout the election cycle. I use data from Intrade [Imai, 2017] and PredictIt [PredictIt] on daily closing prices for the Democratic presidential candidate’s market in a particular state (representing daily predictions of the state-level Democrat vote share) for 2012 and 2016.³⁶ Again estimating separate two-way fixed effects models for 2012 and 2016 and a triple-difference strategy to compare the two elections, I find no significant pre-trends in the triple-differences specification (see Online Appendix Figure 6).

Finally, I use data from the Cooperative Congressional Election Study, which interviews adults on their turnout and vote intentions in the weeks leading up the election [Kuriwaki, 2024]. According to survey documentation, samples are sufficiently large so that survey responses are roughly representative of each state. Therefore, I can compare voters’ intent to turn out and, conditional on intent to vote, intent to vote for the Democratic presidential and U.S. House candidates in treatment and control states. Using the same methods described above, I find no evidence of differential pre-trends among treatment and control states in either year (see Online Appendix Figures 7, 8, and 9). I further address identification concerns with complementary empirical specifications and several robustness checks, introduced below.

³⁵Campaign advertisement data is reported at the Designated Market Area (DMA) level. The state associated with the DMA is identified, but many DMAs cross state lines. I therefore only examine campaign advertisements for down-ballot candidates (Senate, House, and gubernatorial), where the state in which the race took place is identified.

³⁶This data is only available for twenty-one states in 2016, but there is a mix of treatment and control states represented: 12 control and 9 treated.

4.1.3 Results

Table 2 shows results from estimating equation (3). I find no effect of the Comey letter on overall turnout rate. However, I find significant effects of the letter on partisan vote shares: a 3.2 percentage point increase in Republican presidential vote share and a 2 percentage point decrease in Democratic presidential vote share. Table 2 also displays an increase in average Republican down-ballot vote share of 2.9 percentage points and a complementary decrease in average Democratic down-ballot vote share of 3.2 percentage points. In Section 6, I discuss interpretation of these magnitudes.

Table 2: Voters most able to respond to the Comey letter were more likely to vote for Republican candidates in presidential and down-ballot races.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Treat $\times$ Post	0.009 (0.006)	0.008 (0.006)	-0.020* (0.011)	-0.024 (0.015)	0.032*** (0.010)	0.035** (0.015)	-0.032** (0.014)	-0.034* (0.017)	0.029* (0.017)	0.030 (0.018)
<i>N</i>	5782		5782		5782		5400		5400	
Mean	0.560		0.347		0.618		0.370		0.604	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows results from estimating equation (3). I include time-varying controls in the right-hand column under each outcome. I calculate turnout rate as a proportion of voting-age population; presidential vote shares are proportions of total presidential votes cast. I calculate U.S. House, Senate, and gubernatorial vote shares as proportions of total ballots cast in the respective races before averaging the three together to calculate the down-ballot vote share. I drop races without a Democratic or Republican candidate from these averages, leading to the different number of observations for the down-ballot outcomes.

4.1.4 Robustness Checks

Results are robust to the inclusion of time-varying county economic and demographic controls including median income, unemployment rate, population density, median age, number of men per 100 women, percent non-white, percent of those 25 and up with a Bachelor’s degree or higher, and share white without a college degree (right-hand columns under each outcome variable in Table 2). Additionally, results are robust to the inclusion of a term interacting 2016 with share white without a college degree, given the large gap in 2016 among white voters by education [Doherty et al., 2018]: see Online Appendix Table 1.

Note that counties in Florida and Oklahoma may be a superior “treated” group in this

specification, as both Florida and Oklahoma had no-excuse early voting periods that did not open until after 10/28/16. In a specification using only Florida and Oklahoma counties as treated units, both treated and control voters live in states with early voting: the difference is just in when early voting is offered. Results are robust to such a refinement of the design (see Online Appendix Table 2).

Identification also requires that we assume that any late-election information in 2012 did not affect turnout or partisan vote shares. Some may view Hurricane Sandy, which made landfall in the U.S. on October 29th, 2012, as late-election information which benefited President Obama (since his handling of the storm was positively regarded by both parties). In Online Appendix Table 4, I omit counties in Connecticut, New Jersey, and New York, which had the largest populations in storm surge zones during Hurricane Sandy [Center for International Earth Science Information Network, 2012]: results are robust to this omission.

It is possible that this approach picks up larger dynamics within the election, rather than voters' response to the singular shock of the Comey letter. To ensure this is not the case, I conduct placebo tests using three alternative dates to define treatment: 9/26/16 (the first presidential debate), 10/7/16 (Access Hollywood tape release and start of Wikileaks Podesta email releases), and 10/19/16 (the final presidential debate). The results in Table 2 cannot be replicated when defining treatment around any of these dates: results are shown in Online Appendix Table 5. This suggests that the difference-in-differences design identifies the effect of the Comey letter.

A similar concern is that this approach captures differential trends across treatment and control states over a number of elections: perhaps non-early voting states were becoming more right-leaning relative to early voting states. To address this, I conduct two additional placebo tests. Keeping the definition of treatment the same, I re-estimate equation (3) comparing 1) the 2012 and 2008 elections (with 2012 as the "treated" election) and 2) the 2014 and 2018 elections (with 2018 as the "treated" election) using county-level electoral returns from Leip. I find no effects of treatment in either of these specifications (shown in Online Appendix Tables 6 and 7, respectively), lending support to the idea that I am estimating the effect of the Comey letter itself and not broader political trends in treatment and control states.

Finally, in Online Appendix Table 8, I estimate results from Table 2 separately for presidential battleground (Panel A) and non-battleground states (Panel B) using classifications

from [Cook Political Report \[2016a\]](#).³⁷ There is no effect of treatment on states’ battleground status (as shown in Online Appendix Figure 1), but it is plausible that the letter’s effect differed in these two sets of states due to varying levels of campaign activity. However, down-ballot effects of the letter were of importance in both sets of states: nearly one-half of battleground U.S. House races and one-third of battleground Senate races in 2016 took place in non-presidential battleground states [[Ballotpedia](#), [h,g](#)]. I find that down-ballot effects on Democratic and Republican vote shares and the effect on Republican presidential vote share do not significantly differ between the two groups.

4.2 Comparisons of Early and Election Day Voters

4.2.1 Empirical Approach

While my main empirical strategy relies on comparing states with and without early voting, I am also able to compare early voters and Election Day voters *within* early voting states. I study six early voting states (Arkansas, Georgia, Iowa, Louisiana, Maryland, and North Carolina) which report votes cast on Election Day and during the early voting period separately.³⁸ I leverage this disaggregated data to compare votes cast by early voters and Election Day voters in 2012 and 2016, considering Election Day voters as “treated,” as they voted after the letter’s release in 2016. It should be noted that early voting extended past 10/28 in each of these states, and so some of the early voters (“controls”) would have had knowledge of the letter when casting their vote.³⁹ However, I still expect to observe the Comey letter’s effect under the assumption that sufficiently many early voters cast their ballot ahead of 10/28 in 2016, possessing no knowledge of the letter when they did so.

For this reason, I expect to observe larger effects in states that had more (vs. less) days of early voting available ahead of the letter’s release. In states with more days, vote shares from early voting periods should be more representative of voters’ preferences without knowledge of the letter, as it is much more likely that many of them cast their ballot ahead of the letter’s

³⁷Online Appendix Figure 10 displays variation in convenience voting laws for battleground and non-battleground states separately.

³⁸Vote totals were obtained from each state’s elections website [[Arkansas Secretary of State](#), [Georgia Secretary of State](#), [Iowa Secretary of State](#), [Louisiana Secretary of State](#), [Maryland State Board of Elections](#), [North Carolina State Board of Elections](#)].

³⁹In this way, this strategy mirrors how early voting states are considered “partially shielded” from the Comey letter and thus used as controls in my first specification. Here, the analogous group is the early voters from these states.

release (unlike, for example, Maryland, where early voting began on 10/27 in 2016).⁴⁰ Figure 3, which plots the relationship between the number of early voting days ahead of the letter’s release and early voting turnout rate, provides suggestive evidence in favor of this argument.⁴¹ States with more early days of voting have more voters cast their ballot early. While I cannot be sure that a significant portion of these votes were cast prior to the letter’s release, I argue this is a more plausible explanation of the relationship than the alternative: states with more days of pre-10/28 early voting experiencing a differentially large concentration of early voting in the final days ahead of Election Day.

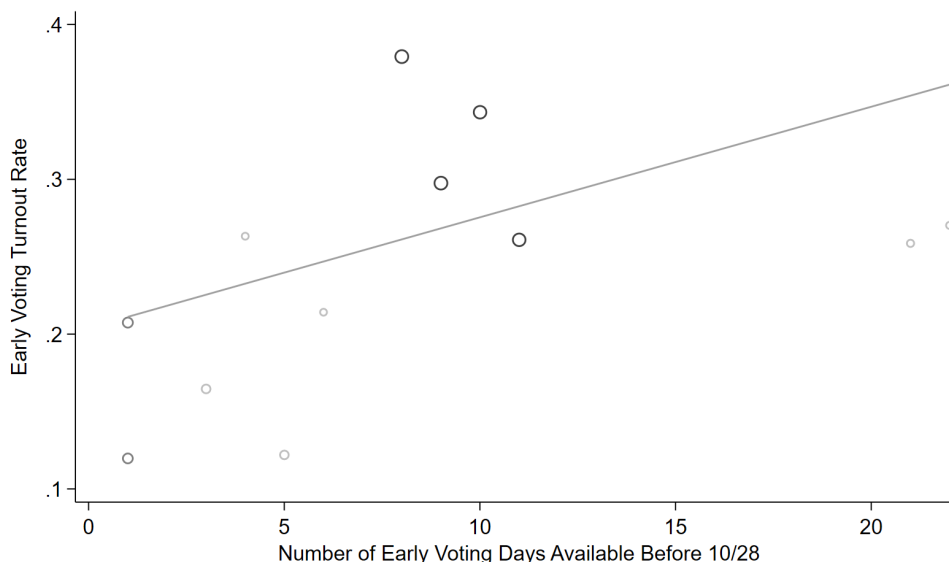


Figure 3: This figure shows a binned scatterplot (weighted by voting-age population) of the number days of early voting available ahead of 10/28 and early voting turnout rate (out of voting-age population) for states used to estimate Table 3. Markers are weighted by voting-age population.

Motivated by the relationship described in Figure 3, I consider a difference-in-differences specification with continuous treatment, leveraging the difference between counties in states with many vs. few days of early voting prior to 10/28. In addition to the traditional parallel trends assumption, this specification requires a further “strong parallel trends” assumption [Callaway et al., 2024]. In particular, I must assume that counties with many days of early

⁴⁰Similarly, Election Day vote shares in these same states are likely more representative of voters actually impacted by the letter. In each state, there was likely a group of voters who strongly preferred Democratic candidates and would vote for them no matter what. These voters are more likely to be captured in the early voting share (rather than Election Day) in states with many days of early voting, given the greater chance to cast their ballot early.

⁴¹I refer to state laws to determine the number of days of early voting prior to 10/28 in each state in 2012 and 2016.

voting prior to 10/28 would have experienced similar effects to counties with few days of early voting prior to 10/28 had they received a “lower dose” (fewer early voting days ahead of the letter’s release) instead. I estimate

$$y_{pcst} = \alpha + \delta T_p + \gamma P_t + \eta F_c + \beta(D_{st} \times T_p \times P_t) + \epsilon_{pcst}, \quad (4)$$

where T_p takes a value of 1 for votes cast on Election Day and 0 for votes cast in early voting periods, and P_t takes a value of 1 in the 2016 election and 0 otherwise. I include county fixed effects, F_c , to account for any time-invariant differences across counties which affect election outcomes. The “dose,” D_{st} , is the number of days early voting was available prior to 10/28. This is determined independently by each state and only enters the specification in 2016.

Note that each county is observed four times: twice in 2012 (early voting period and Election Day) and twice in 2016 (again, early voting period and Election Day). There is variation in election administration across counties, even within the same state [[National Conference of State Legislatures](#)], which may introduce correlation between each of these observations. As such, I cluster at the county level. As in my first specification, I weight regressions by county-year voting-age population. Outcomes of interest are Democratic and Republican presidential vote shares, as well as average vote shares for Democratic and Republican down-ballot candidates. As in equation (3), I average together the Democratic (or Republican) vote shares in U.S. House, Senate, and gubernatorial elections, excluding any races which did not have both a Democratic and Republican candidate.

4.2.2 Results

Table 3 shows results from estimating equation (4). I find that that a county with 10.17 days of early voting ahead of the letter’s release (the average number of days of early voting prior to 10/28 across all states with no-excuse early voting in 2016) saw a decrease of 2.6 percentage points in Democratic presidential vote share and an increase of 1.7 percentage points in Republican presidential vote share in Election Day voters relative to early voters. Effects on down-ballot races are similar: a 2.8 percentage point decline in average Democratic down-ballot vote share and a 1.9 percentage point increase in average Republican down-ballot vote share.

Table 3: Voters most able to respond to the Comey letter were more likely to vote for Republican candidates in presidential and down-ballot races.

	Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
# of Days EV × Treat × Post	-0.0026*** (0.001)	-0.0026*** (0.001)	0.0017*** (0.000)	0.0017*** (0.000)	-0.0028*** (0.001)	-0.0025*** (0.001)	0.0018*** (0.001)	0.0017*** (0.001)
<i>N</i>	2072		2072		1814		1814	
Mean	0.377		0.595		0.383		0.584	
Controls	✓		✓		✓		✓	

Standard errors are clustered at the county level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows results from estimating equation (4). I include time-varying controls in the right-hand column under each outcome. Presidential vote shares are proportions of total presidential votes cast. I calculate U.S. House, Senate, and gubernatorial vote shares as proportions of total ballots cast in the respective races before averaging the three together to calculate the down-ballot vote share. I drop races which did not have both Republican and Democratic candidates running, leading to the different number of observations for the down-ballot outcomes.

Since the treatment “dose” (number of days of early voting pre-10/28) is determined separately by each state, we might consider clustering at the state, rather than county, level. Note that treatment assignment is “partially clustered,” as only half of the units in each state become treated (Election Day vote shares), implying clustering at the state level might lead to overly conservative standard errors [Abadie et al., 2023].⁴² I show the results from equation (4) clustered at the state level in Online Appendix Table 9. Results for the down-ballot partisan vote shares are robust when considering an unweighted regression with an ordinary wild restricted bootstrap, as recommended by MacKinnon et al. [2023]. Statistical precision decreases when estimating weighted regressions. However, I show in Online Appendix Table 10 that the decrease in Democratic presidential vote share and increase in Republican down-ballot vote share are robust to a non-continuous version of equation (4), adding confidence in the results.

I interpret the results in Table 3 as secondary evidence in support of my main findings in Table 2, where I find exposure to the letter decreased Democratic presidential and down-ballot vote shares by 2 and 3.2 p.p., respectively, and increased Republican presidential and down-ballot vote shares by 3.2 and 2.9 p.p., respectively. These two empirical strategies leverage similar, but separate, sources of variation and produce comparable results, while the lower statistical precision in the secondary strategy is expected due to a smaller sample size of only 6 states. Both sets of estimates also may represent underestimates of the Comey letter’s effect, as both strategies include some voters who cast their ballot after the letter’s

⁴²See Online Appendix Section 2.7 for discussion.

release as controls. In the first specification, these are voters in early voting states who either voted during the early voting period but after 10/28 or on Election Day. In the second, these are early voters who voted after 10/28.

5 Mechanisms

5.1 Comparisons of Pre- and Post-Comey Letter Early Voters

5.1.1 Empirical Approach

In the first two empirical designs, control voters are only partially shielded from the Comey letter: early voting periods extended past October 28th, so early voters (or voters in early voting states) are simply considered “more constrained” in their ability to respond to the letter, as a significant amount of them would have already voted. To my knowledge, no state reports votes received by each candidate at a finer level of disaggregation than the one leveraged in Section 4.2. However, North Carolina reports daily counts of early votes cast by registered party [North Carolina State Board of Elections]. I use this disaggregated data on partisan early voting turnout to compare pre-October 28th early voters to post-October 28th early voters in 2012 versus 2016, as only the latter group was able to incorporate information from Comey’s letter into their vote in 2016.

The structure of North Carolina’s early voting turnout data grants me the ability to make within-state comparisons and compare early voters to other early voters. That is, this method rules out confounding cross-state and cross-voting method differences. Additionally, studying turnout by party registration provides insight into the mechanisms discussed in Section 3. Thus far, I have been unable to distinguish between turnout and vote-switching. I find no effect of the Comey letter on overall turnout rate in Table 2. In theory, this could be caused by either 1) offsetting increases in turnout by moderate right-leaning voters and decreases in turnout by moderate left-leaning voters, or 2) vote-switching among moderate left-leaning voters, while turnout remains unchanged. However, a decrease in turnout by registered Democrats would indicate changes along the extensive, rather than intensive, margin. Although they come at the cost of examining only one state, these are valuable contributions.

I estimate the following two-way fixed effects event study:

$$y_{cdt} = \delta_d + \delta_t + \sum_{k=-19}^{-13} \mu_k \mathbf{1}\{d = k\} \times treat_t + \sum_{k=-11}^{-3} \gamma_k \mathbf{1}\{d = k\} \times treat_t + \epsilon_{cdt}, \quad (5)$$

where c indexes county, d indexes days relative to Election Day, and t indicates year. Note

that $treat_t$ is a dummy variable equaling 1 in 2016 and 0 in 2012. The letter was released eleven days ahead of Election Day, so I omit the day before. As in the first two empirical strategies, I weight by county-year voting-age population from SEER [National Bureau of Economic Research (NBER)]. Outcomes of interest are turnout by party registrants (i.e., voters registered as Democrats, etc.) as a share of total early votes.

It should be noted that ahead of the 2016 election, a North Carolina election reform law (passed in 2013) was partially stricken down by a federal appellate court, leaving counties with the ability to change hours and locations for early in-person voting. The majority of counties expanded early in-person hours, but 17 counties chose to restrict them. Using a border-county analysis, Walker et al. [2019] find no evidence of uniform effects on turnout from changes to early in-person total hours, evening hours, or Saturday hours, nor from changes to number of early in-person voting sites. They do find effects for some subgroups of voters in response to changes in the availability of Sunday early in-person voting, so I omit Sunday voting from the above analysis. I cluster at the county-level to account for these changes.

All counties continued to offer early voting each weekday beginning 19 days ahead of Election Day. While the director of the North Carolina Republican Party encouraged county elections board members to limit the availability of early voting, many of those same officials did not support such plans [Campbell, 2016]. Given the push from a top Republican official to limit early voting, however, we might expect that counter-mobilization efforts led to higher turnout from Democrats in the first few days of early voting. With these caveats, there is no reason to expect that the changes to early in-person voting would have had differential effects for Democrat versus Republican voters beginning on the ninth day of early voting, which is what I test for with equation (5).

5.1.2 Results

Figure 4 displays results from estimating equation (5). I find a 2.2 p.p. decrease in registered Democrats' early turnout share and a corresponding 2.2 p.p. increase in registered Republicans' early turnout share (as a proportion of all voters, including those not registered as a Republican or Democrat) following the letter's release.⁴³ Note that we observe the strongest effects immediately after the letter's release, but there are no offsetting effects later on. That is, it is not the case that registered Democrats were deterred initially but made up for this

⁴³The letter was dominating news headlines by 2:00 PM eastern on October 28th [Silver, 2017], so that several hours of early voting were left on the day of the letter's release. For example, Wake County, North Carolina's most populous county, had early voting sites open until 7:00 PM that day (<http://msweb03.co.wake.nc.us/bordelec/downloads/14earlyvoting/info/EarlyVotingSitesWithTurnout.pdf>).

drop in turnout a few days later. These findings are consistent with the turnout mechanism discussed in Section 3; i.e., they suggest that the letter caused a decline in turnout among left-leaning voters who likely would have supported Clinton (and her down-ballot co-partisans), generating coattail effects.

Note that Figure 4 exhibits pre-trends. As theorized above, there may have been an initial increase in Democratic early voting share in 2016 in response to perceived attempts by Republican officials to curtail early voting. Presumably, this would have led to a gradual decline in Democratic early turnout share over the early voting period as there was a fixed supply of registered Democrats to mobilize. I test the sensitivity of the average effect over the first two post-periods under smoothness restrictions as in [Rambachan and Roth \[2023\]](#) in Online Appendix Figure 11. I find that the effect is robust to a linear extrapolation of the pre-trend; however, it would break down under sufficiently large changes in the slope of the existing pre-trend between days.

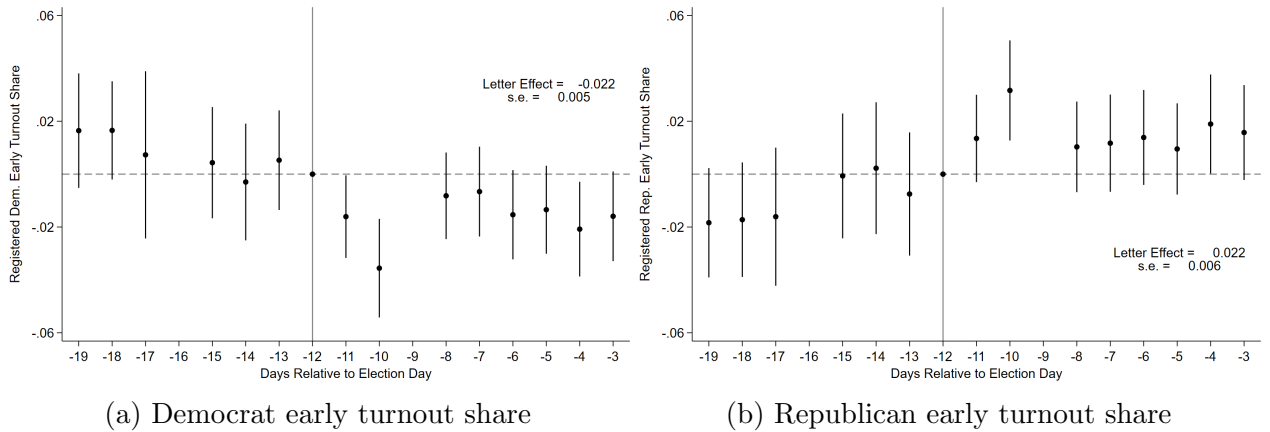


Figure 4: After the Comey letter, the daily share of ballots cast by Democrats (resp., Republicans) in North Carolina decreased (resp., increased).

Notes: This figure displays results from estimating equation (5). The letter was released 11 days ahead of Election Day, so I omit day -12. Standard errors are clustered at the county level; the graph depicts 95% confidence intervals. I omit Sunday voting, explaining the gaps at -16 and -9 days relative to Election Day.

In Online Appendix Table 11, I estimate a two-way fixed effects specification using this data (Panel A1). I use data from [Walker et al. \[2019\]](#) to show that results are robust to controlling for county-year early voting characteristics (number of early voting sites, total number of early voting hours, number of evening early voting hours, number of Saturday early voting hours, and number of Sunday early voting hours) in Panel A2. I also demonstrate that results are robust to the omission of the first three days of early voting (Panel A3), where potential counter-mobilization efforts appear to have increased Democratic turnout. In addition, I provide further suggestive evidence of the turnout mechanism in Panel B by

leveraging weekly partisan early turnout data in Nevada in a similar design, finding a decrease in turnout by registered Democrats after 10/28 in 2016 [Nevada Secretary of State].⁴⁴

5.2 Comparisons of Pre- and Post-Comey Letter Survey Respondents

Survey data from the Cooperative Congressional Election Study (CCES) provides further evidence that is suggestive of the turnout mechanism [Kuriwaki, 2024]. The CCES conducts interviews of over 50,000 American adults throughout the election cycle in each election year, including a question on intent to turn out. Figure 5 documents binned averages of CCES respondents’ turnout intentions, reported separately for registered Democrats and registered Republicans.⁴⁵

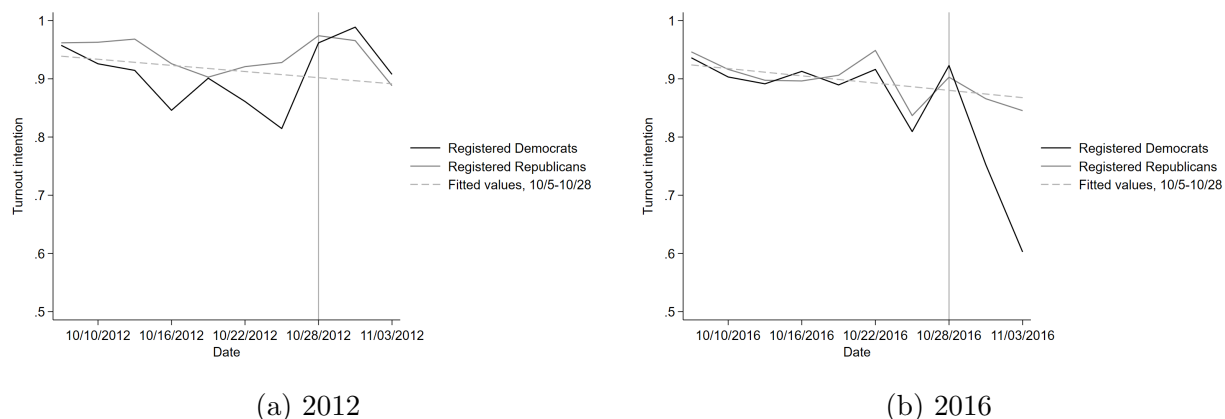


Figure 5: This figure shows CCES respondents’ average intended turnout rate among registered Democrats and Republicans in 2012 and 2016, relative to a trend line based on responses recorded between 10/5 and 10/28.

Notes: I restrict to respondents whose party identification has been validated using voter files. I collapse responses into three-day bins, with the values on the x -axis representing the final day of each bin. The average is calculated using CCES survey weights. I omit respondents who indicate that they have already voted early, as their responses do not include the date they voted, making it impossible to determine whether a respondent surveyed after 10/28 who reports having voted early cast her ballot before or after 10/28.

In 2016, there is a decline in intended turnout among registered Democrats following the date of the letter’s release – a sharper decline than we would have expected based on the trend in responses recorded between October 5th and October 28th. The steep decline is not

⁴⁴Note that this finding is sensitive to the type of bootstrap employed when performing statistical inference; I discuss this in the Online Appendix.

⁴⁵The CCES matches survey respondents’ personal information to public voter files in order to validate their party registration. I use the validated sample, rather than self-reported partisan identity, for a measure of partisan identity that is likely unaffected by the letter’s release. Due to small sample sizes on particular days, I collapse responses into three-day bins: for example, the average turnout intention plotted for 10/28/2016 represents a three-day average across October 26th and October 28th.

present for registered Republicans, nor is it present in 2012 for either party. As I theorize uninformed voters are more likely to be low-propensity voters, these turnout effects may have been concentrated among straight-ticket voters, generating spillover effects for down-ballot Democrats through the turnout mechanism described in Section 3.

Motivated by Figure 5, I estimate individual-level logit regressions using responses from the CCES. I compare respondents' propensity to report intending to turn out before and after 10/28 in 2016, relative to 2012. Results are reported in Online Appendix Table 12. I find a significant decrease in the probability of intending to turn out among registered Democrats and not registered Republicans. Taken together, the results shown in Figure 4, as well as the additional results from weekly partisan early turnout data in Nevada (Online Appendix Table 11) and survey data on turnout intent from the CCES (Online Appendix Table 12), suggest that the Comey letter generated coattail effects for down-ballot Democrats through decreases in turnout among left-leaning voters.

6 Interpreting Magnitudes: An Instrumented Difference-in-Differences Approach

In Tables 2 and 3, I find that exposure to the Comey letter led to significant decreases in down-ballot Democratic vote shares (and increases in their Republican counterparts). These findings are suggestive of presidential coattail effects: the popularity of a presidential candidate affected the performance of her party's candidates in concurrent races. Scholars have long sought to estimate presidential coattail effects. However, this is often a difficult task: rarely can the popularity of a presidential candidate be separated from the general attitudes toward her political party.

In order to estimate presidential coattail effects in my setting, I turn to an instrumented difference-in-differences (DDIV) model [Hudson et al., 2017, Duflo, 2001]. This model scales the difference-in-differences effect of the Comey letter on down-ballot partisan vote shares by its effect on partisan presidential vote shares. In other words, it allows me to obtain the effect of a one percentage point increase in partisan presidential vote share, rather than the effect of exposure to the Comey letter itself. The DDIV coefficient β is estimated via the following instrumental variables system:

$$y_{ct} = \alpha + \delta T_c + \gamma P_t + \beta \widehat{Y}_{ct} + \epsilon_{ct} \quad (6)$$

$$Y_{ct} = \theta + \rho T_c + \pi P_t + \eta(T_c \times P_t) + \mu_{ct}, \quad (7)$$

where T_c and P_t are defined as in equation (3). Here, exposure to the Comey letter instruments for Y_{ct} , the Republican presidential vote share in county c in year t .^{46,47} In the second-stage equation, outcomes of interest y_{ct} include average vote shares for Republican and Democratic down-ballot candidates. I weight by voting-age population in both estimations.

Note that the first-stage parameter, η from equation (7), is equivalent to the coefficient estimated by equation (3) when regressing on presidential vote shares.⁴⁸ The DDIV coefficient β can be written as the ratio of the reduced-form and first-stage parameters. The reduced-form parameter can be found by estimating:

$$y_{ct} = \nu + \tau T_c + \xi P_t + \phi(T_c \times P_t) + \omega_{ct}, \quad (8)$$

i.e. equation (3) when regressing on down-ballot candidates' vote shares. As discussed in Hudson et al. [2017], the DDIV design requires several assumptions. Just as in the difference-in-differences model estimated in equation (3), the assumption of parallel trends must hold. Similar to traditional instrumental variables models, monotonicity and the exclusion restriction must hold. In my context, this requires assuming 1) exposure to the Comey letter did not cause an increase in Democratic presidential vote share for some subpopulations and 2) knowledge of the Comey letter did not affect partisan down-ballot vote shares except through its effect on the popularity of each party's presidential candidates.

Table 4 presents the results from equations (6) and (7) where the outcomes of interest are down-ballot partisan vote shares (averaged over U.S. House, Senate, and gubernatorial elections). Results indicate that a one percentage point increase in the Republican presidential vote share leads to a 0.79 percentage point increase in the down-ballot Republican vote share.⁴⁹

⁴⁶I instrument for Republican presidential vote share rather than Democratic for two reasons: a stronger first-stage and ease of interpretation (exposure to the Comey letter increased the Republican – not Democratic – presidential vote share).

⁴⁷In the formal treatment of DDIV provided in Hudson et al. [2017], Y_{ct} is assumed to be “a discretely- and positively-valued treatment.” However, the authors note that it is straightforward to extend to a setting with continuous treatment – such as a presidential vote share.

⁴⁸For the IV exercise, I drop all counties which only had one major party run in all down-ballot races. This results in a sample of 5400, rather than 5782, observations, and means that η from equation (7) and β from equation (3) differ slightly.

⁴⁹Note that much of the coattails literature is dedicated to estimating the extent of spillovers between presidential and U.S. House races. In Online Appendix Table 13, I re-estimate equations (6) and (7) using partisan U.S. House vote shares as the sole outcomes of interest.

Table 4: An increase in the Republican presidential vote share leads to concurrent increases (decreases, resp.) in Republican (Democratic, resp.) down-ballot vote shares.

	(1)	(2)
Rep. Down-Ballot Vote Share	0.787** (0.334)	0.786** (0.325)
Dem. Down-Ballot Vote Share	-0.872*** (0.311)	-0.880*** (0.308)
First-Stage (Rep. Pres. Vote Share)	0.037*** (0.014)	0.038** (0.016)
N	5400	5400
Controls		✓

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows results from an instrumental variables system. The first-stage is given by equation (7), and the second-stage is given by equation (6). In column (2), I include time-varying controls. I drop observations where only one major party competed in all relevant races.

7 Conclusion

This paper is the first to make use of an October surprise to estimate presidential coattail effects: using an instrumented difference-in-differences design, I find that a one percentage point increase in Republican presidential vote share leads to a 0.79 percentage point increase in average Republican down-ballot vote share. Results in this paper indicate that James Comey’s letter to Congress, released just eleven days ahead of the 2016 presidential election, increased both the presidential and down-ballot vote shares of Republican candidates. Using cross-state variation in the availability of early voting, I find that exposure to the letter led to an increase in the Republican presidential vote share of 3.2 percentage points – larger than the eight closest state margins that election.

Complementary empirical designs suggest that the effect of the letter is robust to limiting to both within-state and early voting state-only comparisons. In a specification comparing partisan vote shares among early voters and Election Day voters in early voting states, I find evidence suggestive of a differential increase in Republican presidential and average down-ballot vote shares in counties with more early voting days (vs. less) ahead of the letter’s release. I also provide evidence on the mechanisms of coattail effects: when comparing pre-10/28 early voters to post-10/28 early voters, I find that exposure to the letter led to a 2.2 percentage point decrease in registered Democrats’ early turnout share in North Carolina.

This is consistent with the theory that negative information regarding a presidential candidate can demobilize low-propensity, uninformed voters who likely would have voted for her down-ballot co-partisans had they turned out.

The findings of this paper indicate that the popularity or electoral strength of a party's presidential candidate has meaningful downstream effects. These spillovers have important policy implications: namely, coattail effects increase the likelihood of unified, rather than divided, government. In leading to more occurrences of unified government, coattail effects may impact legislative efficacy and the types of policies passed. Coattail effects raise numerous strategic considerations for down-ballot political candidates, too: for example, how closely to align with prominent co-partisan candidates or even when to run for office. Finally, the finding that late-election information regarding a presidential candidate has down-ballot effects suggests that there are strong incentives for the strategic release of information during presidential campaigns beyond first-order effects on the presidential race.

Online Appendix

1 Support for Identification

1.1 Balance Test

In equation (3), I leverage cross-state variation in convenience voting laws at the time of the Comey letter’s release. Counties in states with early voting open prior to 10/28 are controls, and counties in states 1) with early voting open only *after* 10/28 and 2) with no early voting at all are treated. The identifying assumption is the parallel trends assumption: absent the Comey letter, treatment and control counties should have trended similarly in the outcomes of interest. However, one might be concerned that states with and without early voting may have been experiencing different political trends between 2012 and 2016, violating this assumption.

I conduct a balance test in order to address this concern. I examine whether treatment is related to items on states’ ballots whose presence may reflect either varying political trends or a differential reason to turn out or to vote for a specific partisan candidate in that election year. I focus on four broad categories: number (and type) of ballot initiatives, toss-up Senate or gubernatorial races, the availability of straight-ticket voting, and the presence of a third-party presidential candidate. In addition, I consider states’ battleground status, i.e. whether the presidential race was considered a competitive one in that state.

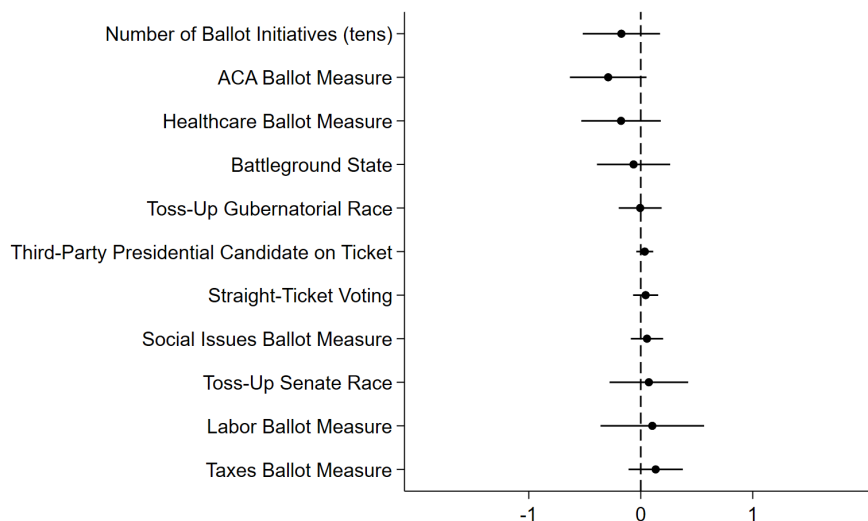
Ballot initiatives have been shown to increase voter turnout [Tolbert et al., 2001]; for this reason, I test whether the number of ballot initiatives is related to treatment. Different categories of ballot measures can serve to mobilize specific groups of voters [Garretson, 2014], potentially leading to spillover effects on other election outcomes. For example, Gardner et al. [2025] find that the presence of abortion-related ballot measures boosted Democrats’ performance in congressional races in the 2022 midterm elections. I collect data from Ballotpedia on a few major categories of ballot measures (taxes, healthcare, labor, social issues) to test for a relationship between the presence of different ballot initiative categories and treatment [Ballotpedia, c,b].^{1,2}

¹In the 2012 presidential election, four states had ballot initiatives aimed at restricting the Affordable Care Act. I classify these ballot measures in their own category separate from other healthcare-related initiatives.

²The “social issues” category includes measures on gun rights, same-sex marriage, abortion, the death penalty, and marijuana legalization.

Especially competitive races (either presidential or down-ballot) might increase voter turnout, amplify partisan campaigning efforts, etc. I collect data on which states 1) were considered battleground states (i.e., those with competitive presidential elections) or 2) had toss-up Senate or gubernatorial races in 2012 and 2016, as defined by the Cook Political Report [Cook Political Report, 2016c,b, 2012c,b]. Third-party presidential candidates were particularly important in the 2016 election, so I also test whether the presence of third-party presidential candidates on the ballot can be predicted by treatment.³

Since I leverage exposure to the Comey letter to estimate coattail effects, I am especially concerned with voters' access to straight-ticket voting, which should theoretically increase the strength of such effects (as it makes voting for a single party up and down the ballot particularly easy). In fact, several states did away with straight-ticket voting in the years preceding 2016. I code each state's straight-ticket availability in 2012 and 2016 using data from The National Conference of State Legislatures. I re-estimate equation (3) and place indicator variables on the left-hand side for each of these variables (or for number of ballot measures, number of ballot measures divided by ten). Results are shown in Online Appendix Figure 1. There is no statistically significant relationship between treatment and any of the covariates, supporting the parallel trends assumption.



Online Appendix Figure 1: There is no effect of treatment on states' ballot items or battleground status. That is, early voting and non-early voting states did not appear to be experiencing different political trends between 2012-2016.

Notes: This figure displays results from variations of equation (3) which place ballot characteristics and battleground status on the left-hand side.

³Out of all states in my analysis, only Oklahoma did not feature third-party presidential candidates on their ballot, and only in 2012.

1.2 Comparisons of Treatment and Control States with High-Frequency Data

Identification in equation (3) requires that treatment and control states were trending similarly both before and after the letter’s release date in 2012 and prior to 10/28 in 2016. Several high-frequency datasets allow for various tests of this assumption. In the following section, I use data on campaign expenditures, campaign donations, television campaign advertisements, predicted vote shares for presidential candidates from prediction markets, and survey data on voters’ intentions to compare treatment and control states throughout the weeks leading up to Election Day in 2012 and 2016.

I estimate variations of two equations. First, I estimate a two-way fixed effects event study:

$$y_{sw} = \delta_s + \delta_w + \sum_{k < -3} \mu_k \mathbf{1}\{w = k\} \times treat_s + \sum_{k \geq -2} \gamma_k \mathbf{1}\{w = k\} \times treat_s + \epsilon_{sw}, \quad (9)$$

where s indexes state and w indexes weeks relative to Election Day. Here, $treat_s$ is a dummy variable equal to 1 in states without early voting available ahead of 10/28/16 and 0 otherwise. The Comey letter was released eleven days ahead of Election Day, so the third week prior to Election Day is omitted. I adapt the above specification to the daily level where relevant, omitting the twelfth day ahead of Election Day.⁴ I estimate equation (9) separately for 2012 and 2016.

Second, I estimate the following triple-difference specification:

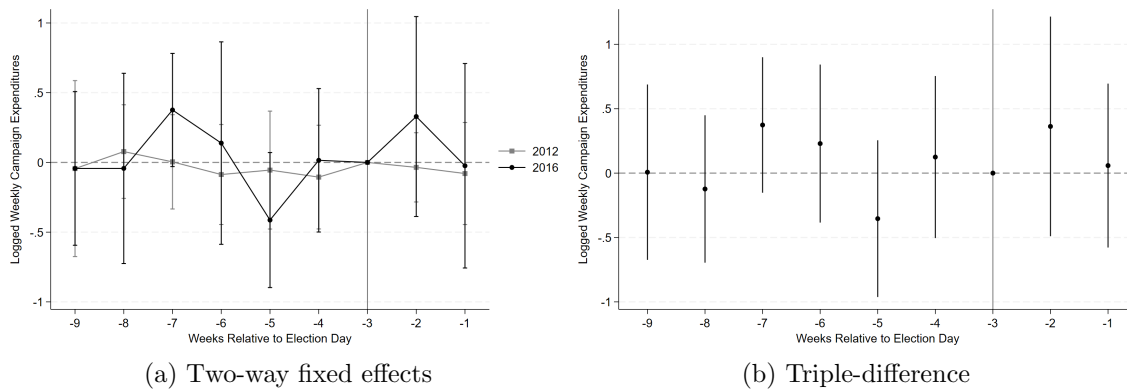
$$y_{swt} = \delta_{st} + \delta_{sw} + \delta_{wt} + \sum_{k < -3} \mu_k \mathbf{1}\{w = k\} \times treat_{st} + \sum_{k \geq -2} \gamma_k \mathbf{1}\{w = k\} \times treat_{st} + \epsilon_{swt}, \quad (10)$$

where s indexes state, w indexes weeks relative to Election Day, and t indexes year. Here, $treat_{st}$ is a dummy variable equal to 1 in states without early voting available ahead of 10/28/16 (only turning on in 2016) and 0 otherwise. Again, I omit the week prior to the Comey letter’s release and adapt to the daily level where relevant.

I first consider measurements of campaign activity in the form of campaign expenditures and campaign donations using data from the Federal Election Commission [[Federal Election Commission](#)]. I collect data on total expenditures from principal campaign committees

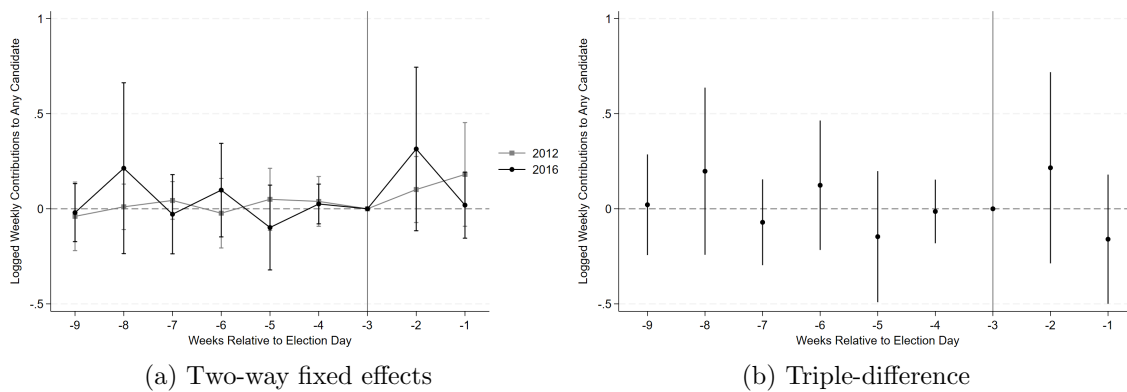
⁴Here, I center estimations relative to Election Day rather than October 28th. Time to Election Day is likely an important determinant of some of these outcomes: in particular, campaign expenditures, donations, and advertisements. For example, we might expect a surge in campaign expenditures in the final week before Election Day, but this week would have different calendar dates in 2016 (Election Day on November 8th) than in 2012 (Election Day on November 6th).

of Republican and Democratic candidates and total donations made to Republican and Democratic candidates. I aggregate these data (available daily at the ZIP-level) to the state-week level, as such a fine level of measurement may introduce quite a bit of noise. I estimate equations (9) and (10), where the outcomes of interest are logged total campaign expenditures and logged total campaign donations.⁵ Results are shown in Online Appendix Figures 2 and 3. While estimates are noisy, there are no significant pre-trends in either year.



Online Appendix Figure 2: Logged weekly campaign expenditures did not trend differently in early voting and non-early voting states in 2012 or 2016.

Notes: This figure displays results from two event study specifications using logged campaign expenditures as the outcome variable. Panel A displays results from estimating equation (9), and Panel B displays results from estimating equation (10). I weight by voting-age population and cluster at the state level; 95% confidence intervals are displayed.

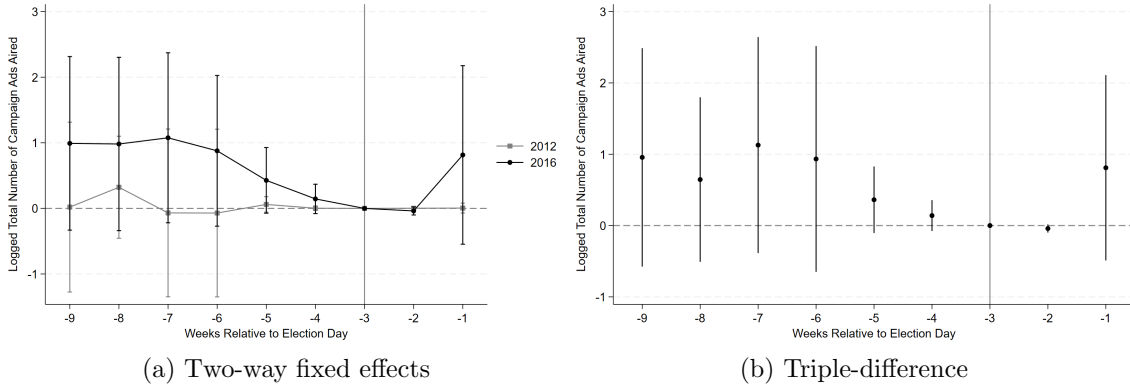


Online Appendix Figure 3: Logged weekly campaign donations did not trend differently in early voting and non-early voting states in 2012 or 2016.

Notes: This figure re-estimates equations used in Online Appendix Figure 2 using logged total campaign donations as the outcome variable. I weight by voting-age population and cluster at the state level; 95% confidence intervals are displayed.

⁵I use $\ln(y + 1.1)$ to account for any zeros in the state-week panel.

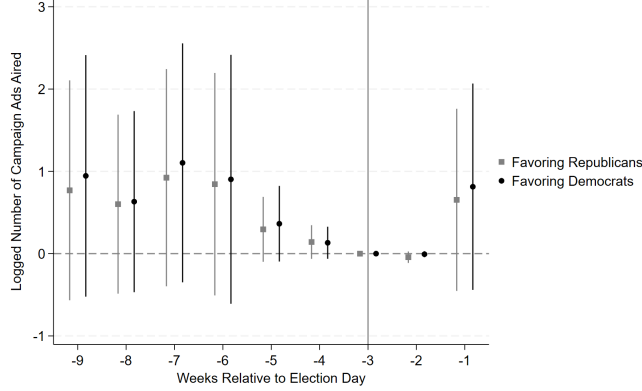
Next, I use daily television campaign advertisement data from the Wesleyan Media Project [Franklin Fowler et al., 2023, Fowler et al., 2023]. Data on total TV campaign ads from Democratic and Republican candidates is reported at the Designated Media Market (DMA) level. The state associated with the DMA is reported, but this is not necessarily the state in which the ad aired, as many DMAs cross state lines. Therefore, I study only advertisements in down-ballot races (Senate, House, and gubernatorial), where the state in which the race took place is identified. I aggregate the data to the state-week level and estimate equations (9) and (10), where the outcome of interest is logged total TV campaign ads.⁶ Results are shown in Online Appendix Figure 4. There were no significant pre-trends in 2012. In 2016, there was increased focus on non-early voting states in the very early weeks of the election, but this did not differ significantly by party (see Online Appendix Figure 5).



Online Appendix Figure 4: Logged weekly television campaign advertisements did not trend differently in early voting and non-early voting states in 2012. In 2016, there was increased focus on *non*-early voting states in the very early weeks of the election, but this did not differ significantly by party.

Notes: This figure re-estimates equations used in Figure 2 using logged number of television campaign advertisements as the outcome variable. I weight by voting-age population and cluster at the state level; 95% confidence intervals are displayed.

⁶Again, I use $\ln(y + 1.1)$ to account for any zeros.



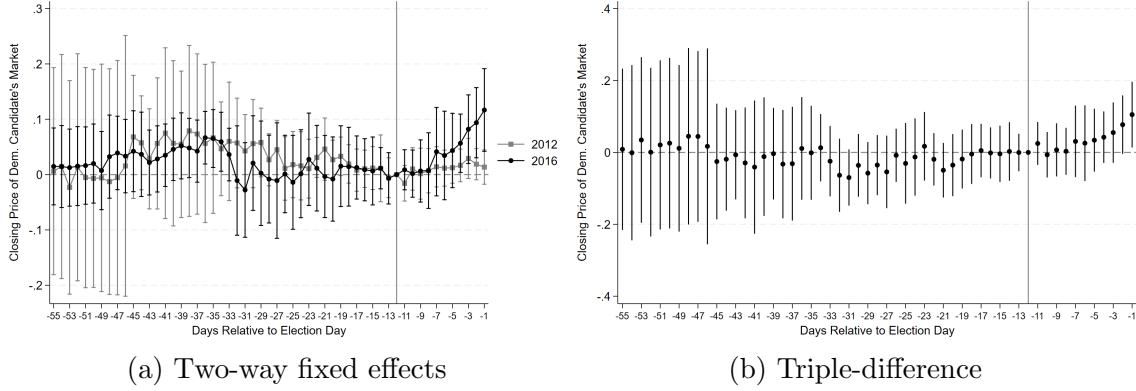
Online Appendix Figure 5: Triple-difference coefficients for logged television campaign advertisements did not differ significantly by party.

Notes: This figure displays the coefficients from the triple-difference event studies for two different outcomes: logged campaign ads favoring Republican candidates and logged campaign ads favoring Democratic candidates.

I next consider daily closing prices for the Democratic presidential candidate’s market in each state (representing daily predictions of the state-level Democrat presidential vote share). This data comes from Intrade for 2012 [Imai, 2017] and PredictIt for 2016.⁷ This data is only available for twenty-one states in 2016, but there is a mix of treatment and control states represented: 12 control and 9 treated.⁸ I estimate equations (9) and (10), where the outcome of interest is the closing price for the Democratic presidential candidate’s market (i.e., predicted Democrat presidential vote share). Results are shown in Online Appendix Figure 6. There are no significant pre-trends in the triple-difference specification. Interestingly, there is an upward trend in predicted Democrat presidential vote share for treated states after the letter’s release (even though I find these groups saw a decrease in Democrat presidential vote share using election outcomes). This is likely due to the composition of states for which this data is available.

⁷Source: <https://www.predictit.org/>.

⁸The control states represented are Arizona, Georgia, Iowa, Indiana, Maine, North Carolina, Nevada, Ohio, Texas, Utah, Wisconsin, and Wyoming, and the treated states are Florida, Michigan, Missouri, Mississippi, New Hampshire, New York, Pennsylvania, South Carolina, and Virginia.

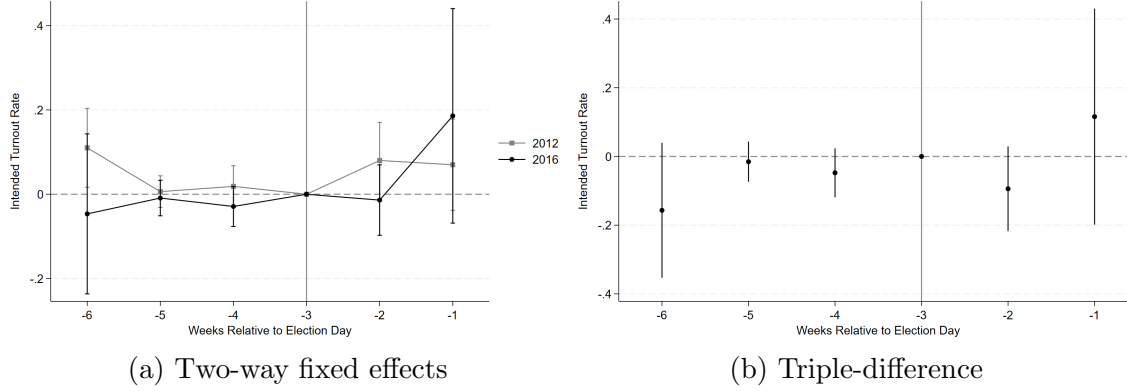


Online Appendix Figure 6: There are no significant pre-trends in daily closing price (predicted state-level vote share) of the Democratic presidential candidate's market in 2012 or 2016.

Notes: This figure displays results from two event study specifications. In Panel A, two-way fixed effects models are estimated separately in 2012 and 2016 and displayed together. In Panel B, a triple-difference specification is estimated. Panel A shows results from estimating $y_{sd} = \delta_s + \delta_d + \sum_{k=-55}^{-13} \mu_k \mathbf{1}\{d = k\} \times treat_s + \sum_{k=-11}^{-1} \gamma_k \mathbf{1}\{d = k\} \times treat_s + \epsilon_{sd}$, where s indexes state and d indexes days relative to Election Day. Note that $treat_s$ is a dummy variable equaling 1 in states without early voting open prior to 10/28/16 and 0 otherwise. Panel B shows results from estimating $y_{sdt} = \delta_{st} + \delta_{sd} + \delta_{dt} + \sum_{k=-55}^{-13} \mu_k \mathbf{1}\{d = k\} \times treat_{st} + \sum_{k=-11}^{-1} \gamma_k \mathbf{1}\{d = k\} \times treat_{st} + \epsilon_{sdt}$, where s indexes state, d indexes days relative to Election Day, and t indexes year. Note that $treat_{st}$ is a dummy variable equaling 1 in states without early voting open prior to 10/28/16 (only turning on in 2016) and 0 otherwise. Each specification is weighted by voting-age population with standard errors clustered at the state level; 95% confidence intervals are displayed. Note that this data is only available for a subset of 21 states in 2016 (9 treatment and 12 control); I thus limit the sample in 2012 to match.

Finally, I use daily survey data from the Cooperative Congressional Election Study (CCES), a large-scale survey which interviews adults on their turnout and vote intentions throughout the election cycle [Kuriwaki, 2024]. According to survey documentation, samples are sufficiently large so that survey responses are roughly representative of each state. I aggregate the responses to the state-week level using CCES survey weights and examine intentions to turn out, and conditional on turnout intention, intentions to vote for the Democratic presidential and U.S. House candidates.

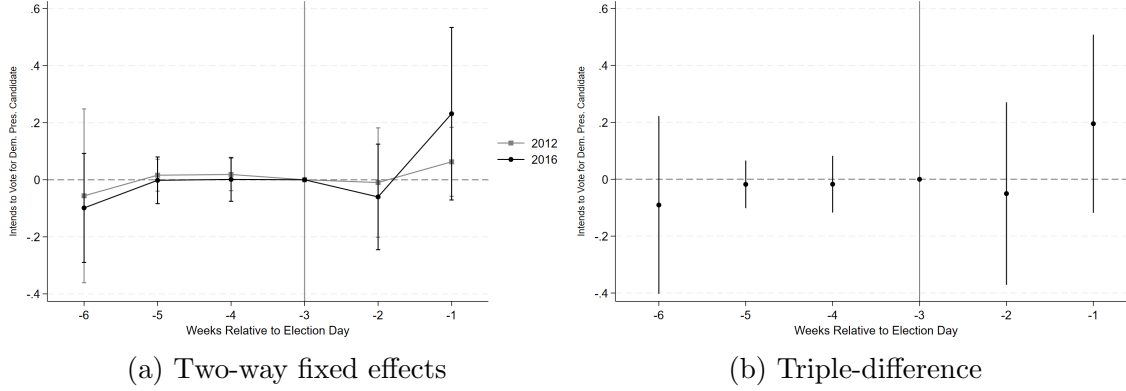
Results are presented in Online Appendix Figures 7, 8, and 9. There are no significant pre-trends in 2012 and 2016. Interestingly, there is a (non-significant) increase in turnout intention, as well as intention to vote Democrat, in non-early voting states in 2016. It is important to note that all of these responses are conditional on not yet having turned out. Therefore, we would expect higher turnout intentions in states where no voters have had the opportunity to vote yet.



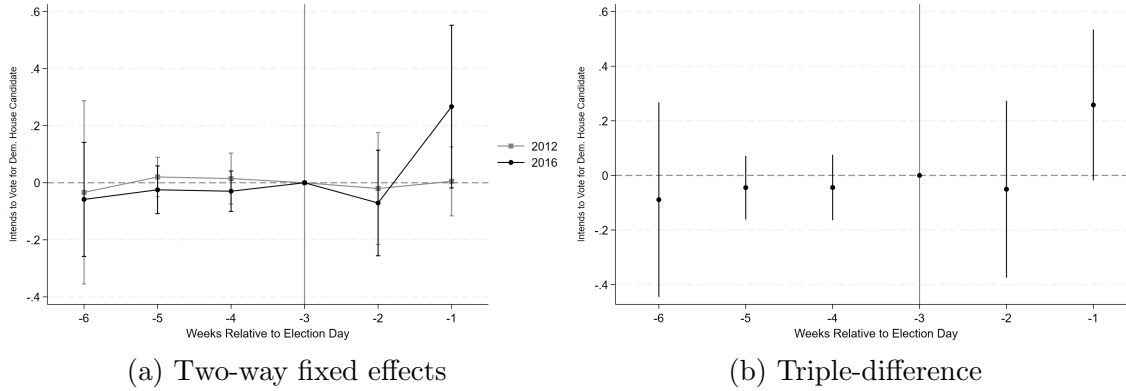
Online Appendix Figure 7: Logged weekly CCES intended turnout rate did not trend differently in early voting and non-early voting states in 2012 or 2016.

Notes: This figure displays results from two event study specifications using logged weekly average intended turnout rates as the outcome variable. Panel A displays results from estimating equation (9), and Panel B displays results from estimating equation (10). Weekly average intended turnout rates are calculated for each state using CCES' survey weights so that they are representative of each state. Standard errors are clustered at the state level; 95% confidence intervals are displayed.

Regarding the (non-significant) increase in intention to vote for Democratic candidates after the letter's release, shown in Online Appendix Figures 8 and 9, it is likely true that each state had a group of voters strongly committed to voting for Democratic candidates. In early voting states, many of those voters may have voted early and therefore would not be represented in the survey in later weeks. In non-early voting states, these voters have not yet had the opportunity to vote, and thus would be represented in the survey in later weeks. Therefore, I interpret these non-significant increases as an indication that marginal voters who had not yet voted in treatment and control states responded to the letter similarly. The difference in state-level responses, then, comes from the fact that voters strongly committed to the Democratic party are no longer in the survey group in control states (as they had already voted early), but are in the survey group in treated states (as they had no chance to vote early).



Online Appendix Figure 8: Logged weekly CCES intended Democratic presidential vote share did not trend differently in early voting and non-early voting states in 2012 or 2016. Notes: This figure re-estimates equations used in Online Appendix Figure 7 using the share of voters who indicated an intention to vote for the Democratic presidential candidate as the outcome variable.



Online Appendix Figure 9: Logged weekly CCES intended Democratic U.S. House vote share did not trend differently in early voting and non-early voting states in 2012 or 2016. Notes: This figure re-estimates equations used in Online Appendix Figure 7 using the share of voters who indicated an intention to vote for the Democratic U.S. House candidate as the outcome variable. I drop races without a Democrat or Republican.

2 Robustness Checks and Additional Specifications

2.1 Accounting for White Voters' Education Status

Given the large gap in 2016 among white voters by education [Doherty et al., 2018], it is important to ensure that equation (3) is not capturing a change among white non-college voters and attributing it to the Comey letter (this would be a concern, for example, if non-early voting states had a higher concentration of white voters without a college degree). In Table 1, I show that treatment and control states have similar shares of white adults without

a college degree in 2012 (in fact, early voting states have a slightly larger share). I include share white without a college degree in my vector of time-varying controls in Tables 2 and 3. Finally, I re-estimate equation (3) with the inclusion of a term interacting 2016 and share white without a college degree. Results are robust to the inclusion of this term, as shown in Online Appendix Table 1.

Online Appendix Table 1: Results are robust to the inclusion of a term interacting 2016 and share white without a college degree.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Treat $\times$ Post	0.008 (0.006)	0.008 (0.006)	-0.024 (0.020)	-0.024 (0.015)	0.036* (0.020)	0.036** (0.015)	-0.035** (0.015)	-0.034** (0.015)	0.032* (0.017)	0.031* (0.016)
<i>N</i>	5782		5782		5782		5400		5400	
Mean	0.560		0.347		0.618		0.370		0.604	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates results from Table 2 after including a term interacting 2016 and share white without a college degree. County-level measures of this variable are from the ACS 5-year estimates [U.S. Census Bureau, a].

2.2 Florida and Oklahoma

In this section, I make a refinement to the design used to estimate Table 2 so that I only compare early voting states to other early voting states. I limit treatment counties to those with early voting open only after 10/28 (all counties in Florida and Oklahoma). Then, I re-estimate (3) with the refined treatment group and same control group (all counties in states with early voting open prior to 10/28). This allows me to make direct comparisons between early voting states only, rather than relying on comparisons between early voting and non-early voting states.

Results are shown in Online Appendix Table 2. Here, there is a change in the overall turnout rate: voters in Florida and Oklahoma, who were more able to respond to the Comey letter, saw a 1.5 percentage point increase in turnout. This is a bit puzzling, as I find that the Comey letter led to a decrease in turnout among registered Democrats. Findings remain qualitatively similar when limiting the control group to counties in neighboring states

(Arkansas, Georgia, Kansas, New Mexico, and Texas; see Panel B).⁹ However, looking at disaggregated turnout of a subset of these neighbors in Online Appendix Table 3, I show that the Comey letter did not lead to a significant change in the early voting turnout rate or Election Day turnout rate using survey data reported by election administrators from [United States Election Assistance Commission](#).¹⁰ Generally, the turnout result appears noisy and is perhaps due to measurement error in the denominator.¹¹

Returning to Online Appendix Table 2, the results show an increase in Republican presidential and down-ballot vote shares, as well as a decline in Democratic down-ballot vote shares. The magnitude and direction of the results remain consistent when comparing Florida and Oklahoma only to their neighboring states (Panel B), though the small sample size appears to cause a loss in statistical precision. Broadly, the results remain consistent with those found in my main specification and reported in Table 2.

Online Appendix Table 2: Voters in Florida and Oklahoma were more likely to vote for Republican presidential and down-ballot candidates than voters in states with convenience voting open before 10/28/16.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Panel A: All Control States										
Treat×Post	0.015*** (0.004)	0.013** (0.006)	-0.006 (0.010)	-0.008 (0.017)	0.017* (0.010)	0.018 (0.015)	-0.048*** (0.011)	-0.048*** (0.015)	0.051*** (0.016)	0.051*** (0.015)
N	4180		4180		4180		3814		3814	
Mean	0.555		0.334		0.629		0.354		0.617	
Controls	✓		✓		✓		✓		✓	
Panel B: Neighboring Control States										
Treat×Post	0.012 (0.104)	0.010 (0.472)	-0.030 (0.185)	-0.008 (0.670)	0.031 (0.162)	0.009 (0.512)	-0.063 (0.275)	-0.027 (0.107)	0.079 (0.409)	0.046* (0.067)
N	1540		1540		1540		1288		1288	
Mean	0.507		0.298		0.674		0.317		0.660	
Controls	✓		✓		✓		✓		✓	

In Panel A, standard errors are clustered at the state level and listed in parentheses.

In Panel B, I cluster at the state level and list wild cluster bootstrap p -values in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Notes: This table re-estimates the results in Table 2 using only counties in Florida and Oklahoma as treatment units. Panel B restricts the set of controls to counties in states neighboring Florida and Oklahoma.

⁹Here, due to the small number of clusters (7 states), I use a wild cluster bootstrap with a six-point Webb distribution in Stata using the “boottest” command [Roodman et al., 2019]. This method does not assume normality, so standard errors are not estimated. I instead report p -values in parentheses.

¹⁰Again, I implement a wild cluster bootstrap with a six-point Webb distribution.

¹¹For example, if voting-age population were growing more quickly in Florida and Oklahoma than documented by the SEER measures, this could lead to a false finding of increased turnout rate.

Online Appendix Table 3: The Comey letter led to no significant change in the Election Day or early turnout rates in Florida and Oklahoma counties, as compared to counties in their neighboring states: Arkansas, Georgia, Kansas, and New Mexico.

	Turnout Rate		Early Turnout Rate		Election Day Turnout Rate	
Treat×Post	0.017*	0.019**	0.024	0.034	-0.007	-0.014
	(0.073)	(0.016)	(0.340)	(0.487)	(0.627)	(0.633)
<i>N</i>	1014		1014		1014	
Mean	0.527		0.227		0.300	
Controls	✓		✓		✓	

I cluster at the state level and list wild cluster bootstrap p -values in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table examines the impact of the letter on overall turnout, as well as turnout by voting method, in Florida and Oklahoma and neighboring control states. Note that the control sample differs from that in Online Appendix Table 2 Panel B: counties which do not report turnout by voting method are dropped from the estimation.

2.3 Hurricane Sandy

Equation (3) takes 2012 as a base year for the difference-in-differences specification. Thus, I assume that any late-election information in 2012 did not affect election outcomes. One potential violation of this assumption comes from Hurricane Sandy, a tropical cyclone which made landfall in the United States around October 29, 2012. Many at the time posited the storm would benefit incumbent Democratic President Obama, who was widely praised for his response. However, as found in Hart [2014], the effect of Hurricane Sandy on the election was “variable and small in magnitude.”

To account for any possible impact of Hurricane Sandy, I re-estimate equation (3) after omitting counties in Connecticut, New Jersey, and New York. These three states had the largest populations in storm surge zones during Hurricane Sandy (Center for International Earth Science Information Network [2012]).¹² As such, it is probable that any effect Hurricane Sandy had on political outcomes was largest there. Results are robust to this omission; see Online Appendix Table 4.

¹²Along with Massachusetts, which is already excluded from my analysis.

Online Appendix Table 4: Results are robust to the omission of Connecticut, New Jersey, and New York.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Treat×Post	0.005 (0.007)	0.005 (0.007)	-0.020* (0.012)	-0.025 (0.016)	0.032*** (0.011)	0.036** (0.016)	-0.042*** (0.013)	-0.045*** (0.016)	0.039** (0.017)	0.040** (0.017)
<i>N</i>	5600		5600		5600		5218		5218	
Mean	0.561		0.342		0.623		0.364		0.610	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates the results in Table 2 after omitting counties in Connecticut, New Jersey, and New York from the analysis.

2.4 Placebo Treatment Dates

It is possible that the central empirical design simply captures changing dynamics throughout the campaign leading up to Election Day. For example, October 7th, 2016, was an important day in the 2016 presidential election. Three major events unfolded: *The Washington Post* reported on the 2005 “Access Hollywood” tape, the U.S. government accused Russia of meddling in the election, and WikiLeaks began releasing emails from John Podesta. Therefore, it is possible that equation (3) captures the effects of that day, rather than the Comey letter itself. It could also be the case more generally that Clinton and the Democratic party would have performed worse with “later” versus “earlier” voters, even in the absence of the letter.

To address this concern, I conduct placebo tests where I redefine treatment around a few key dates during the election cycle: September 26th, the date of the first presidential debate, October 7th, and October 19th, 2016, the date of the final presidential debate. If I find that voters reacted to the placebo “treatment”, then it is likely that the results in Table 2 are reflective of voter preferences shifting over time in response to a number of factors, rather than the singular Comey letter “shock.” As can be seen in Online Appendix Table 5, I do not find that voters responded to these false treatments, increasing confidence in the attribution of my results to the Comey letter.

Online Appendix Table 5: The results in Table 2 do not hold when treatment is redefined around other major dates in the 2016 election cycle (9/26/16, 10/7/16, 10/19/16), suggesting the results were due to the Comey letter and not simply the larger dynamics of the election.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
First Debate	0.010 (0.019)	0.012 (0.018)	0.022 (0.014)	0.020 (0.021)	-0.023** (0.009)	-0.021 (0.016)	0.014 (0.012)	0.010 (0.014)	-0.007 (0.016)	-0.004 (0.017)
Oct. 7, 2016	0.001 (0.013)	-0.000 (0.012)	0.021 (0.014)	0.023 (0.019)	-0.016 (0.014)	-0.020 (0.018)	0.007 (0.019)	0.008 (0.022)	-0.000 (0.020)	-0.001 (0.022)
Final Debate	0.001 (0.006)	0.005 (0.007)	-0.006 (0.016)	-0.014 (0.021)	0.011 (0.016)	0.020 (0.021)	-0.002 (0.019)	-0.012 (0.023)	-0.000 (0.022)	0.009 (0.023)
<i>N</i>	5782		5782		5782		5400		5400	
Mean	0.560		0.347		0.618		0.370		0.604	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates the results in Table 2 after redefining treatment: counties without early voting open before 9/26/16 (“First Debate”), 10/7/16 (“Access Hollywood” tape, WikiLeaks, etc.), 10/19/16 (“Final Debate”) are considered treated in each respective regression. Those with early voting open ahead of those dates are controls.

2.5 Placebo Elections

Similarly, one might be concerned that my main empirical approach captures differential trends across treatment and control states over a number of elections. That is, perhaps it is the case that non-early voting states were becoming more right-leaning throughout the 2010s. To address this, I conduct two additional placebo tests. Keeping the definition of treatment the same, I re-estimate equation (3) comparing 1) the 2012 and 2008 elections (with 2012 as the “treated” election) and 2) the 2014 and 2018 elections (with 2018 as the “treated” election).¹³ Results are shown in Online Appendix Tables 6 and 7. I find no effects of treatment in either of these specifications, lending support to the idea that I am estimating the effect of the Comey letter itself and not broader political trends in treatment and control states.

¹³Note that Washington, D.C., Rhode Island, and Washington all made changes to their early voting laws between 2008 and 2012. I omit them from the first analysis. Connecticut and one New Mexico county are dropped in the second analysis due to missing demographic control data. I omit Colorado, Massachusetts, Minnesota, and Montana in both analyses for consistency with the original specification.

Online Appendix Table 6: The results in Table 2 cannot be reproduced when comparing the 2008 and 2012 elections, suggesting the results were due to the Comey letter and not broader trends over several elections in the 2010s.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Treat×Post	-0.002 (0.006)	-0.001 (0.005)	0.005 (0.007)	0.007 (0.005)	-0.003 (0.008)	-0.004 (0.005)	0.016 (0.018)	0.018 (0.017)	-0.009 (0.017)	-0.011 (0.016)
<i>N</i>	5676		5676		5676		5436		5436	
Mean	0.567		0.395		0.589		0.424		0.559	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates the specification used in Table 2, using 2012 as the “post” election and 2008 as the “pre” election. I drop observations without both Democratic and Republican candidates running, leading to the different number of observations for the down-ballot outcomes. For controls taken from the ACS 5-year estimates, I use data from 2009 for observations in 2008, as 2008 5-year estimates are unavailable.

Online Appendix Table 7: The results in Table 2 cannot be reproduced when comparing the 2014 and 2018 elections, suggesting the results were due to the Comey letter and not broader trends over several elections in the 2010s.

	Turnout Rate		Dem. House Vote Share		Rep. House Vote Share		Dem. Senate Vote Share		Rep. Senate Vote Share	
Treat×Post	-0.000 (0.014)	-0.002 (0.014)	-0.005 (0.017)	-0.009 (0.016)	0.007 (0.016)	0.011 (0.014)	0.034 (0.059)	-0.006 (0.038)	-0.021 (0.056)	0.019 (0.037)
<i>N</i>	5762		5385		5385		3536		3536	
Mean	0.416		0.346		0.633		0.360		0.605	
Controls	✓		✓		✓		✓		✓	

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates the specification used in Table 2, using 2014 as the “post” election and 2018 as the “pre” election. As presidential elections did not take place in these years, I report the vote shares for U.S. House and Senate races separately, rather than an average of House, Senate, and gubernatorial vote shares. I drop observations without both Democratic and Republican candidates running, leading to the different number of observations for the House outcomes. Since not all states held Senate elections in 2014 and 2018, there are even fewer observations for those outcomes.

2.6 Battleground States

In the following section, I present results from Table 2 separately for battleground and non-battleground states. Here, I categorize all states deemed to have competitive presidential races by the Cook Political Report as battleground states [Cook Political Report, 2016a]. Online Appendix Figure 10 displays variation in convenience voting laws for U.S. states in 2016, broken out by battleground status. Recall that there is no effect of treatment on states’ battleground status (Online Appendix Figure 1), meaning that the results in Table 2 are not driven by differentially competitive presidential races in treatment or control states.



Online Appendix Figure 10: These maps display variation in convenience voting laws for U.S. states in 2016, broken into battleground and non-battleground states. The maps were created using “spmap” and “geo2xy” in Stata [Pisati, 2007, Picard and Steptner, 2017].

It is plausible that the letter’s effect differed in presidential battleground and non-battleground states, as the level of campaign activity in the former is likely much higher. However, this is not to say that coattail effects are of no significance in non-presidential battleground states. Note that in 2016, 12 of 23 battleground U.S. House races and 3 of 9 battleground Senate races were held in non-presidential battleground states [Ballotpedia, g,h]. Coattail effects stemming from changes in turnout and/or vote-switching may have had meaningful effects in these close down-ballot races.

In Online Appendix Table 8, I re-estimate results from Table 2 for 2016 presidential battleground states (Panel A) and non-presidential battleground states (Panel B) separately. Note that the effects on Republican presidential vote share, Democratic down-ballot vote share, and Republican down-ballot vote share are stronger in non-battleground states, but effects on these outcomes do not significantly differ by battleground status.

However, the effects on turnout and Democratic presidential vote share do significantly differ in battleground and non-battleground states. In battleground states, exposure to the letter increased turnout by 3 percentage points and led to a very small (and non-significant) increase in Democratic presidential vote share (overall, the letter led to a non-significant increase in Republican presidential vote margin of 1.3 percentage points in battleground states). One interpretation of these results is that in presidential battleground states, the letter was more likely to mobilize turnout among right-leaning low-propensity voters than it was to discourage turnout among left-leaning low-propensity voters and, perhaps, induced vote-switching away from third-party candidates. This interpretation is consistent with the idea that perceived closeness of an election leads to higher turnout, although evidence is mixed [Bursztyn et al., 2023, Gerber et al., 2020]. While the mechanisms may have differed between battleground and non-battleground states, the down-ballot effects of the Comey letter did not.

Online Appendix Table 8: The results in Table 2 appear to be driven by non-battleground states, although the difference between the coefficients for Republican presidential vote share and partisan down-ballot vote shares are not statistically significant.

	Turnout Rate		Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Panel A: Battleground States										
Treat×Post	0.030*** (0.007)	0.029** (0.011)	0.003 (0.015)	0.003 (0.013)	0.016 (0.020)	0.015 (0.019)	-0.020 (0.018)	-0.019 (0.017)	0.016 (0.021)	0.016 (0.020)
N	1664		1664		1664		1538		1538	
Mean	0.590		0.362		0.604		0.376		0.601	
Controls	✓		✓		✓		✓		✓	
Panel B: Non-Battleground States										
Treat×Post	-0.001 (0.008)	-0.002 (0.008)	-0.031** (0.011)	-0.037* (0.019)	0.039*** (0.011)	0.046** (0.019)	-0.037** (0.014)	-0.041** (0.019)	0.033* (0.018)	0.038* (0.020)
N	4118		4118		4118		3862		3862	
Mean	0.548		0.341		0.624		0.368		0.606	
Controls	✓		✓		✓		✓		✓	
Difference in Coefficients										
p-values	0.003	0.024	0.077	0.075	0.291	0.250	0.469	0.377	0.524	0.428

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates results from Table 2 using 2016 battleground states (Panel A) and non-battleground states (Panel B) only. The final row in the table gives the p -values from testing if the coefficients differ significantly using the Z -statistic $Z = \frac{(\beta_a - \beta_b)}{\sqrt{SE(\beta_a)^2 + SE(\beta_b)^2}}$ [Clogg et al., 1995].

2.7 Clustering Dose-Response Specification at the State Level

In this section, I show results from estimating equation (4) and clustering at the state, rather than county, level. The treatment “dose,” i.e., the number of days of early voting available before 10/28, is determined at the state level. Thus, one might consider clustering at that level. Note that there are only six states in this analysis, as (to my knowledge) only six states report vote shares separately for early voting and Election Day periods in my years of interest. Due to the very small number of clusters, I use a wild cluster bootstrap with a six-point Webb distribution, estimated with the command “boottest” in Stata [Roodman et al., 2019]. MacKinnon et al. [2023] note that in the case of very few treated clusters, the wild cluster bootstrap tends to *under*-reject the null. They recommend using the ordinary wild restricted bootstrap in this case. As such, I report both wild cluster bootstrap p -values (in parentheses) and ordinary wild restricted bootstrap p -values (in square brackets) in the table below: in each case, the latter is smaller.

The results are shown in Online Appendix Table 9. I report the results of regressions weighted by voting-age population in Panel A (as in Table 3). I also report unweighted regressions in Panel B, because the recommendation from MacKinnon et al. [2023] to use an ordinary wild restricted bootstrap in the case of few treated clusters pertains to unweighted regressions: MacKinnon [2019] notes that cluster-robust inference for weighted regressions has been understudied in the literature. Results in Table 3 lose statistical precision when clustering at the state level. However, the results for down-ballot vote shares are robust when considering unweighted regressions and an ordinary wild restricted bootstrap.

The loss in statistical significance after clustering at the state level may add some caution in the interpretation of the results in Table 3. However, note that we have what is referred to as “partially clustered assignment” by Abadie et al. [2023], in which case clustering at the state level may be overly conservative. In each state, only half of the units observed are “treated” in the “post” period (the 2016 election): counties’ Election Day vote shares. Counties’ vote shares from early voting periods (even in 2016) are not considered treated. Therefore, clustering at the state level is akin to clustering more coarsely than the level of treatment. In the following section, I show that results from Table 3 are robust to a non-continuous version of the specification, adding confidence in the findings.

Online Appendix Table 9: Results in Table 3 are generally not robust to clustering at the state level. Down-ballot vote share results are robust when considering an unweighted regression with an ordinary wild bootstrap.

	Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Panel A: Weighted by 18+ Population								
Dose×Treat×Post	-0.0026 (0.720) [0.502]	-0.0026 (0.687) [0.466]	0.0017 (0.760) [0.593]	0.0017 (0.743) [0.579]	-0.0028 (0.713) [0.435]	-0.0025 (0.777) [0.470]	0.0018 (0.829) [0.585]	0.0017 (0.807) [0.531]
Panel B: Unweighted								
Dose×Treat×Post	-0.0045 (0.591) [0.224]	-0.0042 (0.531) [0.174]	0.0037 (0.598) [0.251]	0.0035 (0.578) [0.214]	-0.0040 (0.433) [0.075]	-0.0038 (0.242) [0.058]	0.0035 (0.456) [0.073]	0.0033 (0.245) [0.064]
N	2072		2072		1794		1794	
Mean	0.377		0.595		0.383		0.584	
Controls	✓		✓		✓		✓	

I cluster at the state level and list wild cluster bootstrap *p*-values in parentheses, along with ordinary wild bootstrap *p*-values in square brackets.

Notes: This table re-estimates results from Table 3 with regressions clustered at the state, rather than county, level. Panel A reports results when weighting each regression by voting-age population, and Panel B reports results from the unweighted regressions.

2.8 Non-Continuous Specification

In this section, I estimate a non-continuous version of equation (4). In an attempt to ensure that a sizable proportion of the “controls” (early voters) in 2016 cast their ballots ahead of the letter’s release, I limit the sample to states with more than 5 days of early voting ahead of October 28th, as motivated by the high early voting turnout rates in those states, shown in Figure 3. Further, I weight regressions by total votes cast in the presidential race, as I expect counties with more votes cast during the early voting period (versus less) to be more representative of voters’ preferences pre-letter. As in equation (4), I cluster at the county level. I find a significant decrease in Democratic presidential vote share and a significant increase in average Republican down-ballot vote share.

Online Appendix Table 10: Results in Table 3 are somewhat robust to a non-continuous specification weighted by total votes cast.

	Dem. Presidential Vote Share		Rep. Presidential Vote Share		Dem. Down-Ballot Vote Share		Rep. Down-Ballot Vote Share	
Treat $\times$ Post	-0.013** (0.005)	-0.009** (0.004)	0.001 (0.005)	-0.001 (0.004)	-0.007 (0.007)	-0.003 (0.006)	0.012** (0.006)	0.008 (0.005)
N	1432		1432		1262		1262	
Mean	0.385		0.588		0.394		0.579	
Controls	✓		✓		✓		✓	

Standard errors are clustered at the county level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table shows results from a county-period level difference-in-differences specification, weighted by total votes cast in the presidential race: $y_{psct} = \alpha + \delta T_p + \gamma P_t + \eta F_c + \beta(T_p \times P_t) + \epsilon_{psct}$, where T_p takes a value of 1 in the Election Day voting period and 0 in the early voting period, and P_t takes a value of 1 in 2016 and 0 in 2012. County fixed effects are given by F_c . I include time-varying controls in the right-hand column under each outcome: unemployment rate, median income, population density, median age, number of men per 100 women, percent non-white, percent of those 25 and up with a Bachelor’s degree or higher, and share white without a college degree. Presidential vote shares are proportions of total presidential votes cast. I calculate U.S. House, Senate, and gubernatorial vote shares as proportions of total ballots cast in the respective races before averaging the three together to calculate the down-ballot vote share. I drop races uncontested by a major party, leading to the different number of observations for the down-ballot outcomes. Only counties with at least five early voting days occurring before the letter’s release are included.

2.9 Partisan Turnout

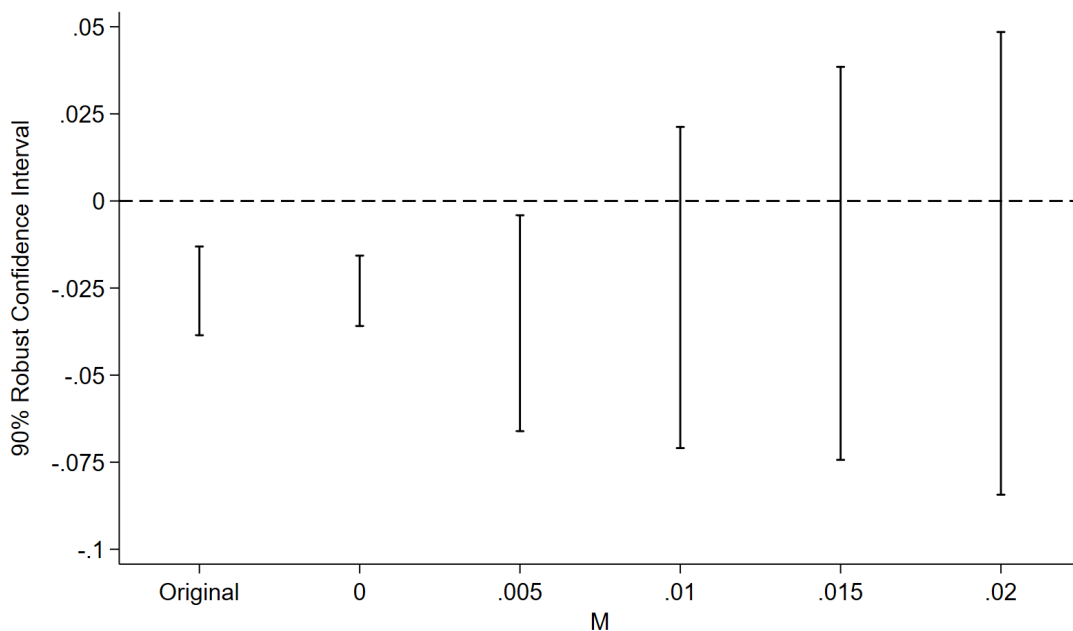
2.8.1 Parallel Trends Sensitivity Analysis: North Carolina

Figure 4 exhibits pre-trends. As discussed in Section 5.1, counties in North Carolina had wider discretion over hours and locations for early in-person voting in 2012 and 2016. A top Republican official in the state encouraged county elections board members to limit the availability of early voting. Although most county election officials did not comply [Campbell, 2016], there may have been counter-mobilization efforts from Democrats in response, leading to higher-than-normal Democratic turnout in the first few days of early voting in 2016. If this were the case, we would expect a gradual decline in Democratic early voting turnout share over the early voting period as the initial boost wore off. In other words, registered Democrats may have front-loaded early turnout in response to perceived threats to early voting.

Under this assumption, I conduct a sensitivity analysis under smoothness restrictions as proposed in Rambachan and Roth [2023]. This allows pre-trends in Democratic early turnout share to continue smoothly past 10/28, ruling out abrupt changes in the slope. I test the

robustness of the average effect over the first two post-periods in Figure 4a (i.e., the day of the letter’s release and the day just following), as these periods are when we see the largest effects.

Online Appendix Figure 11 displays the results of the sensitivity test. The average effect over the first two post-periods is robust to allowing for linear violations of parallel trends ($M = 0$). That is, the decline in Democratic early voting share in the first two post-periods is robust to a linear extrapolation of the pre-trend. The breakdown in robustness occurs between $M = 0.005$ and 0.01 , meaning that the slope of the existing pre-trend would need to change by between 0.5 and 1 percentage points between days to negate the effect.



Online Appendix Figure 11: Sensitivity test under smoothness restrictions

Notes: This figure displays results of a sensitivity test under smoothness restrictions as in [Rambachan and Roth \[2023\]](#). I test the robustness of the average effect over the first two post-periods in Figure 4a.

2.8.2 Two-Way Fixed Effects Estimations: North Carolina and Nevada

Although there is no evidence of systemic effects on turnout in North Carolina stemming from changes in total hours or number of sites [[Walker et al., 2019](#)] and no reason to expect these changes would have differential effects for Democrats and Republicans around the day of the letter’s release, one may be concerned that results in Figure 4 are attributable to potential changes in early voting availability, or Democratic counter-mobilization efforts in response to efforts to reduce early voting availability, rather than the Comey letter.

Here, I estimate the following two-way fixed effects specification using daily turnout data from North Carolina:

$$y_{cdt} = \delta_d + \delta_t + \beta Letter_{dt} + \epsilon_{cdt}, \quad (11)$$

where c indexes county, d indexes days relative to Election Day, and t indexes year. Note that $Letter_{dt}$ is a dummy variable equaling 1 for observations at most eleven days ahead of Election Day and 0 otherwise (and only turning on in 2016). I weight by voting-age population and cluster at the county-level to account for county-level changes in early voting practices.

Results are reported in Online Appendix Table 11 Panel A1. In Panel A2, I include a vector of county-year early voting characteristics from Walker et al. [2019]. In Panel A3, I omit the first three days of early voting. Results remain significant and qualitatively similar: after the letter’s release, registered Democrats’ share of early turnout declined relative to 2012.

I provide additional evidence to bolster the findings in North Carolina by leveraging disaggregated early turnout data provided by Nevada [Nevada Secretary of State]. In 2016, Nevada had a fourteen-day early voting period that lasted between October 22 and November 4. The state reports partisan turnout during Week 1 (10/22/16-10/28/16) and Week 2 (10/29/16-11/4/16) separately, allowing me to compare “Week 1” and “Week 2” early voters in 2012 and 2016.¹⁴ Note that 10/28 is included in the “control” group here as data is reported at a weekly, not daily, level. As such, the results I find likely represent an underestimate of the letter’s true effect.

Using week-county-year observations, I estimate a difference-in-differences specification in Nevada, comparing Week 1 and Week 2 voters in 2016 and in 2012. The identifying assumption here is that, absent the Comey letter, Week 1 and Week 2 voters would have trended similarly in terms of the outcomes of interest. Again, this design also requires that any late-election information in 2012 did not affect political behavior in Nevada. I estimate

$$y_{wct} = \alpha + \delta T_w + \gamma P_t + \beta(T_w \times P_t) + \delta_c + \epsilon_{wct}, \quad (12)$$

where w indexes week, c indexes county, and t indexes year. Here, T_w takes a value of 1 in Week 2 and 0 otherwise, and P_t takes a value of 1 in 2016 and 0 in 2012. I include county

¹⁴In 2012, Week 1 refers to turnout between 10/20/12 and 10/26/12, and Week 2 refers to turnout between 10/27/12 and 11/2/12. In both 2012 and 2016, early voting begins 18 days ahead of Election Day and concludes 4 days ahead of Election Day.

fixed effects δ_c to account for any time-invariant cross-county differences in early voting administration. Outcomes of interest are turnout by party registrants (i.e., voters registered as Democrats, etc.) as a share of total early votes. I weight by voting-age population and cluster at the county level.

Results are shown in Online Appendix Table 11 Panel B. Due to the small number of clusters (17 counties), I implement a wild cluster bootstrap, estimated in Stata with the “boottest” command [Roodman et al., 2019]. This method does not assume normality and as such, standard errors are not estimated. Instead, I report p -values in parentheses. Note that when there are only a few treated clusters, the wild cluster bootstrap tends to *under*-reject. In such cases, MacKinnon et al. [2023] recommend using the ordinary wild restricted bootstrap. As such, I report p -values from the ordinary wild bootstrap in square brackets as well. I find a decrease of 0.9 percentage points in the share of early ballots cast by registered Democrats. Note that this finding loses statistical precision with the wild cluster bootstrap, but the p -value from the ordinary wild restricted bootstrap is 0.037. This is likely an underestimate of the letter’s effect, as the date of its release is included in the “control” group due to data limitations.

Online Appendix Table 11: After the Comey letter was sent, the share of early ballots cast by registered Democrats in Nevada and North Carolina decreased, and the Republican early voting share in North Carolina increased.

	Registered Dem. Early Turnout Share	Registered Rep. Early Turnout Share	Registered Ind. Early Turnout Share
Panel A1: North Carolina			
Letter	-0.022*** (0.005)	0.022*** (0.006)	-0.000 (0.003)
<i>N</i>	2884	2884	2884
Mean	0.469	0.311	0.220
Panel A2: North Carolina – control for county-year early voting characteristics			
Letter	-0.023*** (0.006)	0.023*** (0.007)	-0.000 (0.002)
<i>N</i>	2884	2884	2884
Mean	0.469	0.311	0.220
Panel A3: North Carolina – first three days omitted			
Letter	-0.015*** (0.004)	0.013** (0.006)	0.002 (0.002)
<i>N</i>	2334	2334	2334
Mean	0.457	0.319	0.224
Panel B: Nevada			
Treat×Post	-0.009 (0.239) [0.037]	0.001 (0.853) [0.939]	0.009 (0.961) [0.390]
<i>N</i>	68	68	68
Mean	0.277	0.532	0.191

In Panel A, standard errors are clustered at the county level and listed in parentheses. In Panel B, I cluster at the county level and list wild cluster *p*-values in parentheses, with ordinary wild bootstrap *p*-values listed in square brackets.

In Panel A: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: Panel A shows estimates from equation (11). In Panel A2, I include a vector of county-level early voting characteristics. In Panel A3, I exclude the first three days of early voting. Panel B shows results from estimating equation (12).

2.8.3 Comparing Turnout Intent Across Pre and Post-10/28 Survey Respondents

To provide further evidence of the turnout mechanism, I leverage data from the Cooperative Congressional Election Study [Kuriwaki, 2024]. Figure 5 shows a decline in intended turnout among registered Democrats after 10/28 in 2016. Such a steep decrease in turnout intent is not present for registered Republicans, nor is it present in 2012 for either party. To formally test for an effect of the letter’s release on intent to turn out, I estimate the following

individual-level logit regression:

$$\log\left(\frac{\mathbb{P}(y_{ict} = 1)}{1 - \mathbb{P}(y_{ict} = 1)}\right) = \alpha + \delta T_c + \gamma P_t + \beta(T_c \times P_t) + \epsilon_{ict}, \quad (13)$$

where T_c takes a value of 1 for respondents in 2016 and 0 for respondents in 2012, and P_t takes a value of 1 for respondents answering on or after 10/28 and 0 for respondents answering pre-10/28. I estimate these regressions separately for validated registered Democrats and Republicans and use individual survey weights from the CCES. I report the average marginal effect of $T_c \times P_t$, so that the coefficients can be interpreted as a percentage point change in the probability of reporting intending to vote, and report robust standard errors.

The turnout intent question has six possible answers: “Yes, definitely,” “Probably,” “I already voted (early or absentee),” “Plan to vote early,” “No,” and “Undecided.” I drop respondents who indicate having already voted early or absentee, as they are not asked which date they cast their ballot. When constructing y_{ict} for equation (13), I code respondents who answered “Yes, definitely” and “Plan to vote early” as 1s and those who answered “Probably,” “No,” or “Undecided” as 0s. Results are shown in Online Appendix Table 12. I find that after the Comey letter’s release, registered Democrats were 19.2 percentage points less likely to report intending to vote. I do not find an effect on the probability of turnout intent among registered Republicans. I take this as additional evidence of the turnout mechanism: knowledge of the letter decreased left-leaning voters’ propensity to turn out, losing votes for Democratic candidates.

Online Appendix Table 12: There was a significant decline in turnout intent after 10/28 relative to 2012 among registered Democrats but not Republicans.

	Registered Democrats	Registered Republicans
Treat × Post	-0.192* (0.108)	-0.017 (0.048)
<i>N</i>	16927	13600
Mean	0.912	0.933

Robust standard errors are listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table estimates equation (13). Coefficients represent average marginal effects and can be interpreted as the effect of the letter’s release on the probability that a respondent reports they intend to vote in the upcoming election. I use individual survey weights from the CCES.

2.10 Instrumented Difference-in-Differences Results on U.S. House Vote Shares

The results in Table 4 demonstrate the impact of a one percentage point increase in Republican presidential vote share on partisan down-ballot vote shares. I calculate these down-ballot vote shares by calculating the average of Republican or Democratic U.S. House, Senate, and gubernatorial vote shares, dropping any races without a Republican or Democratic candidate. However, much of the presidential coattails literature uses only U.S. House vote shares as outcomes – likely because, unlike senators and governors, U.S. House representatives face re-election every two years.

Online Appendix Table 13 presents results from equations (6) and (7) where the outcomes of interest are partisan U.S. House vote shares (also from Leip). I find that a one percentage point increase in Republican presidential vote share leads to a 0.51 percentage point increase in Republican U.S. House vote share. These findings are robust to the inclusion of a vector of time-varying county economic and demographic controls in column (2).

Online Appendix Table 13: An increase in Republican presidential vote share leads to concurrent increases (decreases, resp.) in Republican (Democratic, resp.) U.S. House vote shares.

	(1)	(2)
Rep. U.S. House Vote Share	0.514** (0.254)	0.548** (0.224)
Dem. U.S. House Vote Share	-0.490* (0.291)	-0.527* (0.303)
First-Stage (Rep. Pres. Vote Share)	0.035*** (0.013)	0.037** (0.016)
<i>N</i>	4910	4910
Controls		✓

Standard errors are clustered at the state level and listed in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table re-estimates results from Table 4 using only U.S. House vote shares as the outcome of interest in the second-stage. I drop observations where only one major party competed.

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